

BFS Traversal

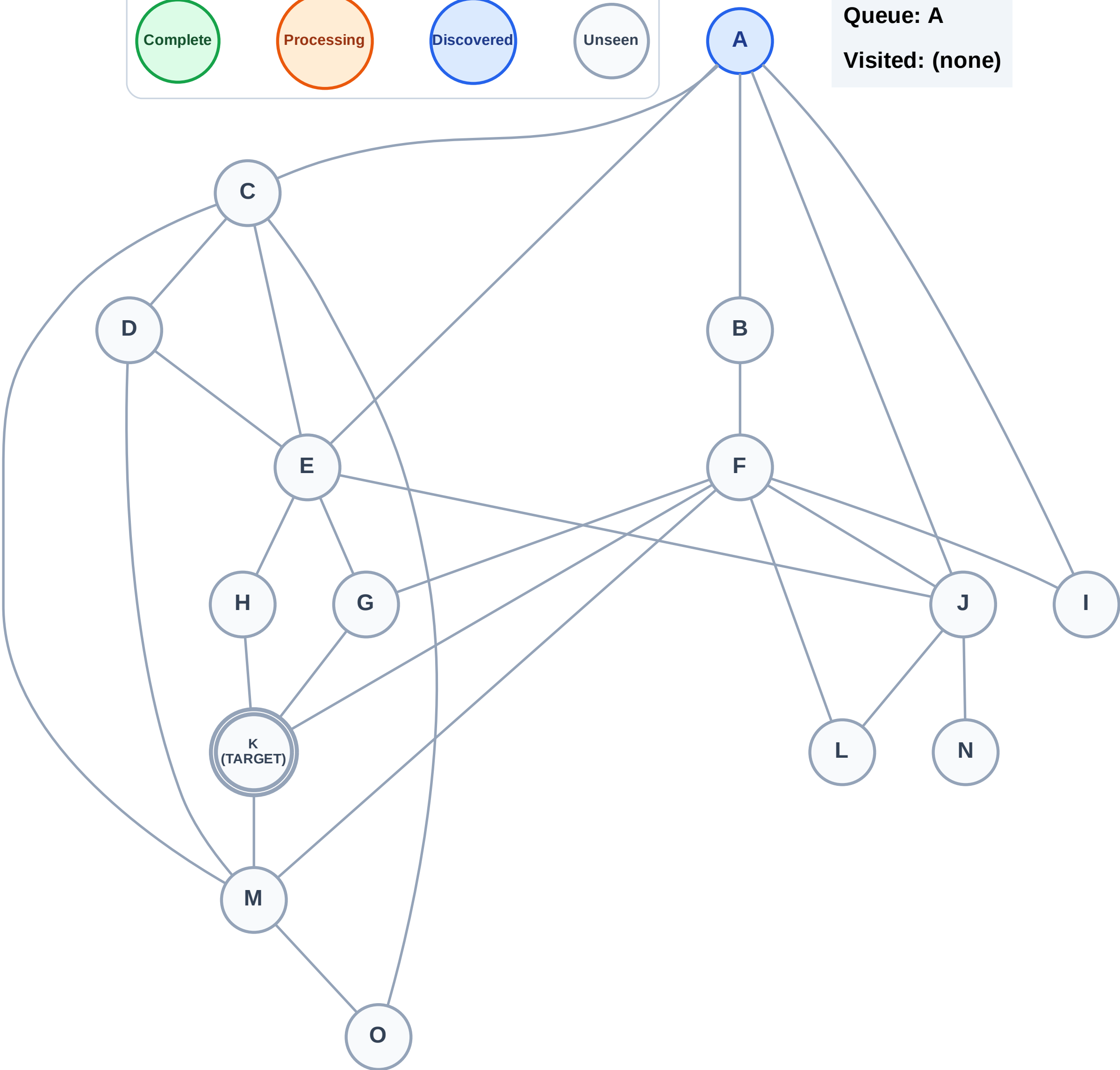
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

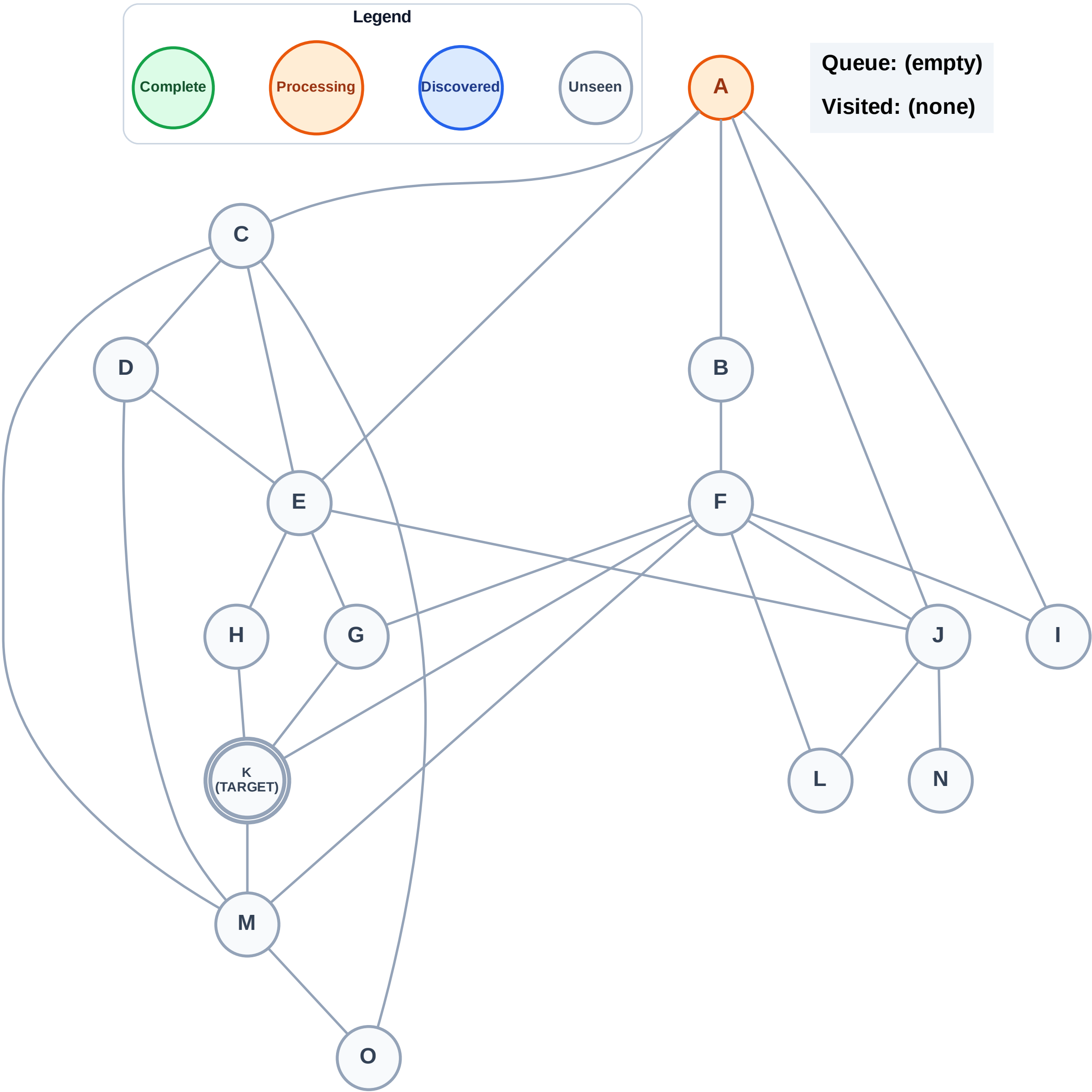
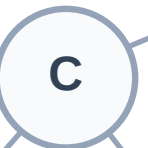
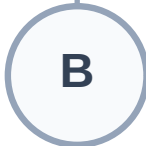
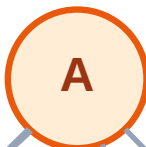
Step 2: Process node A

Legend



Queue: (empty)

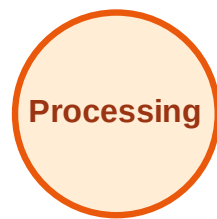
Visited: (none)



BFS Traversal

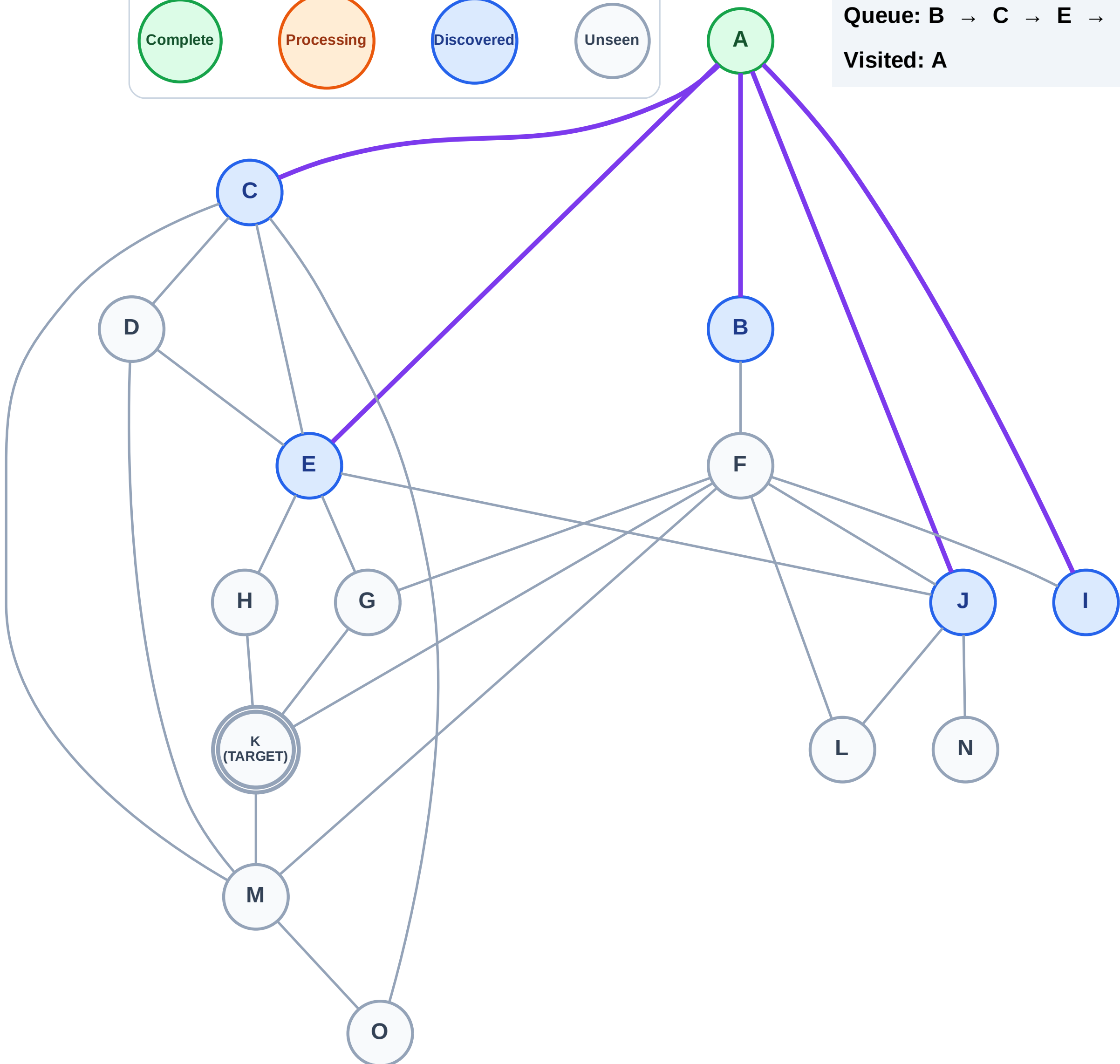
Step 3: Discover B, C, E, I, J from A

Legend



Queue: B → C → E → I → J

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

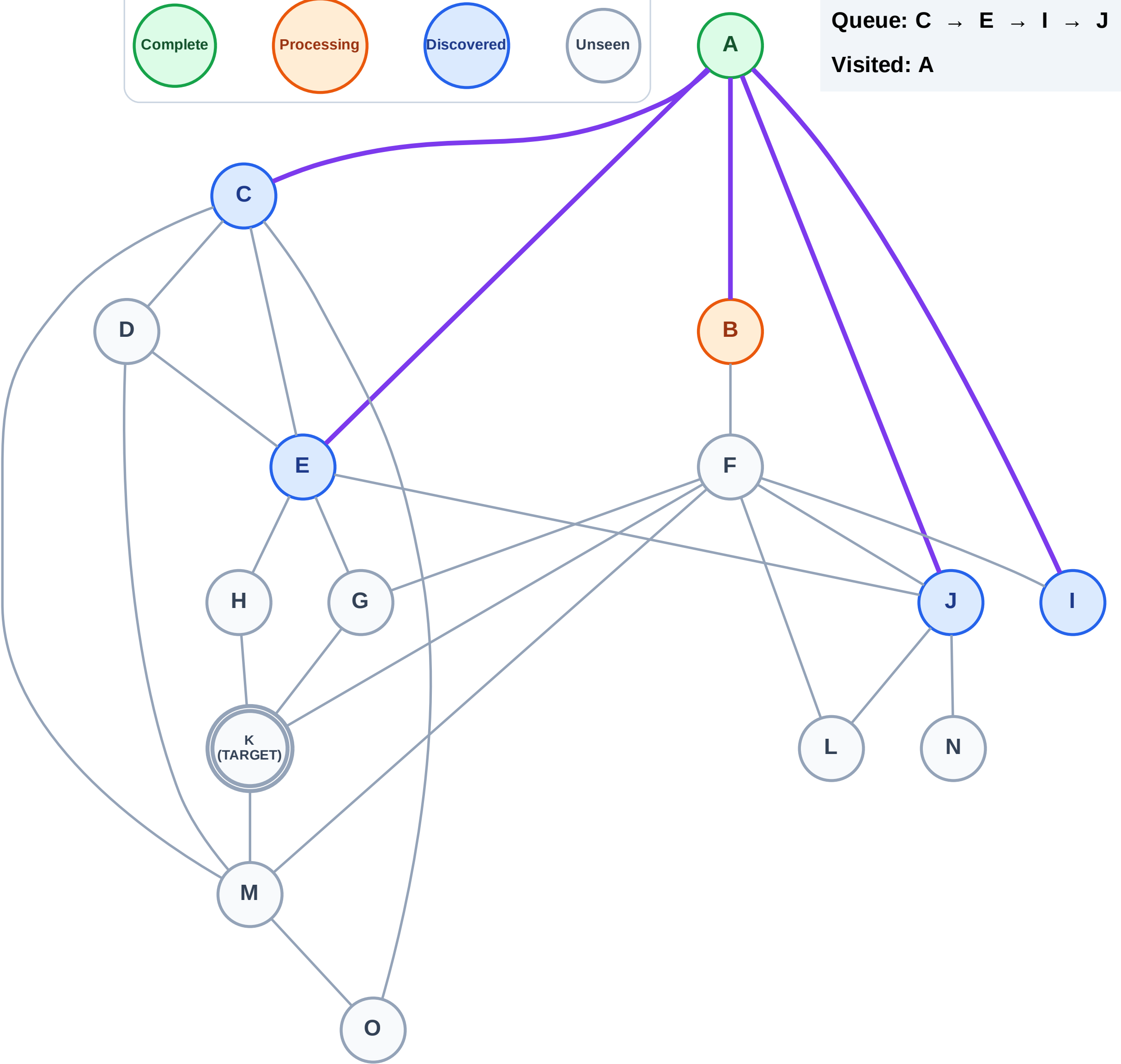
Processing

Discovered

Unseen

Queue: C → E → I → J

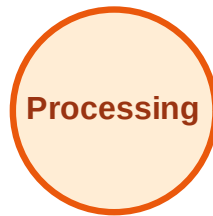
Visited: A



BFS Traversal

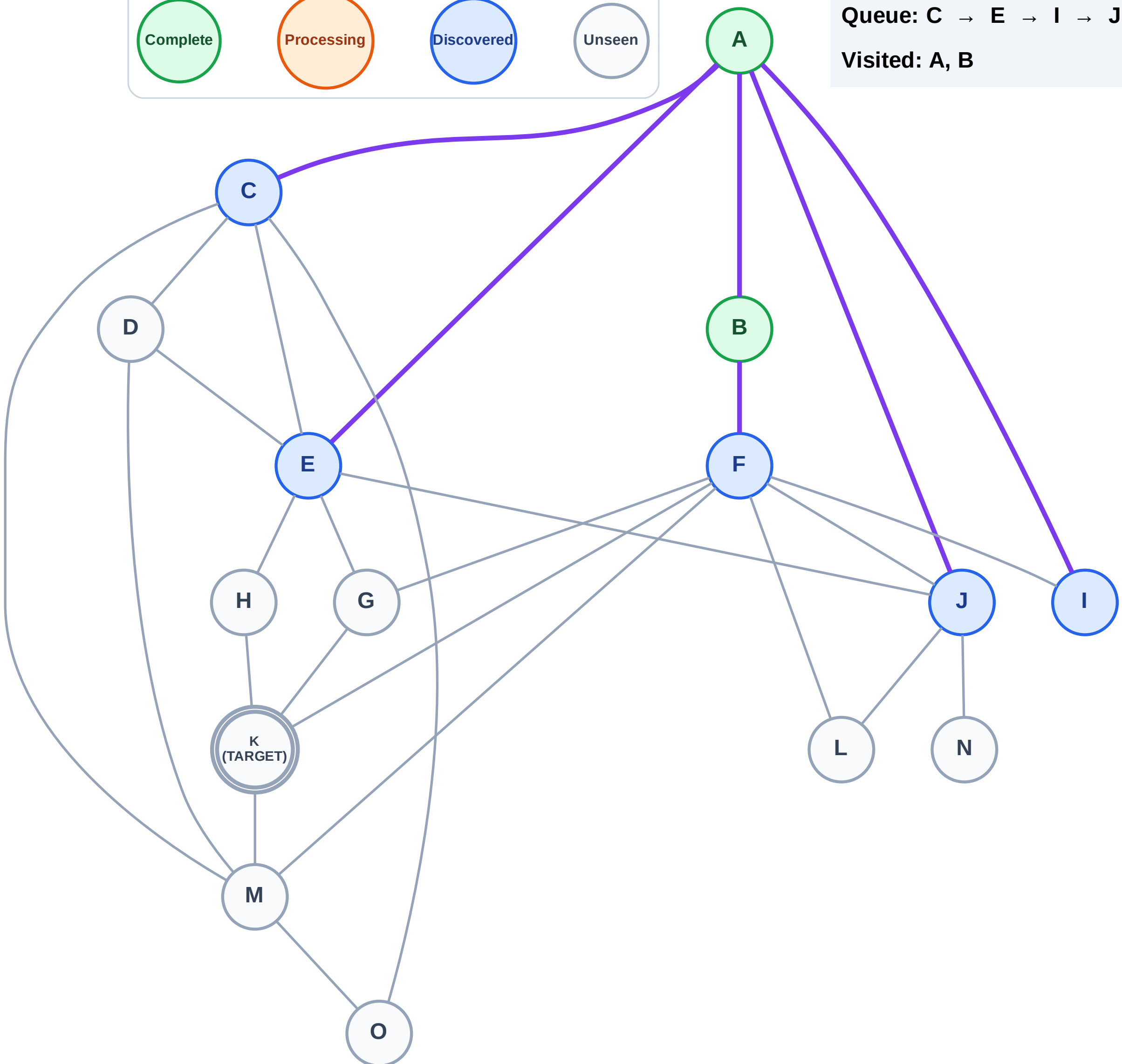
Step 5: Discover F from B

Legend



Queue: C → E → I → J → F

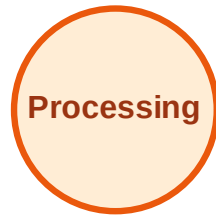
Visited: A, B



BFS Traversal

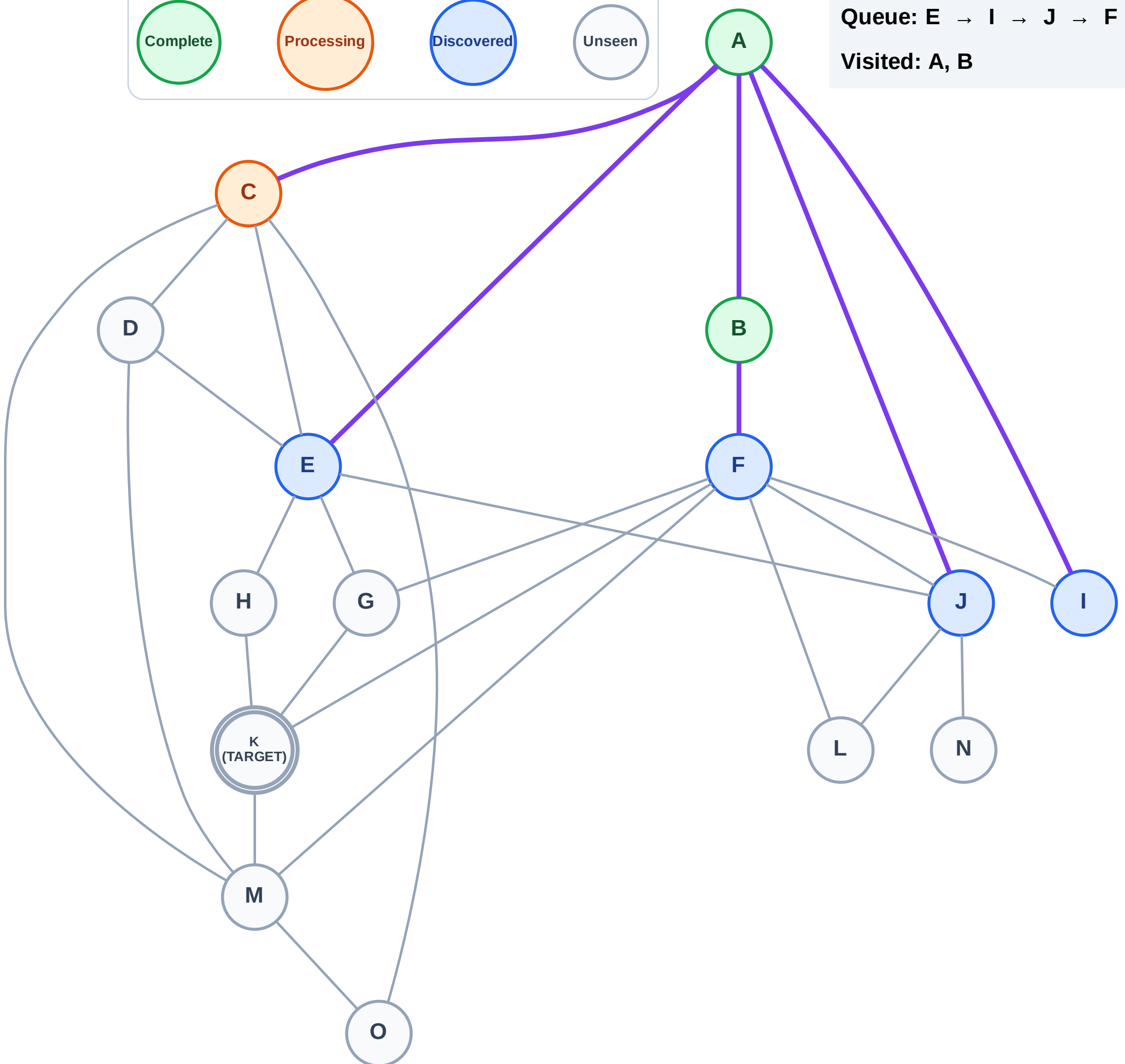
Step 6: Process node C

Legend



Queue: E → I → J → F

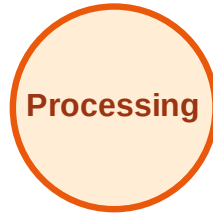
Visited: A, B



BFS Traversal

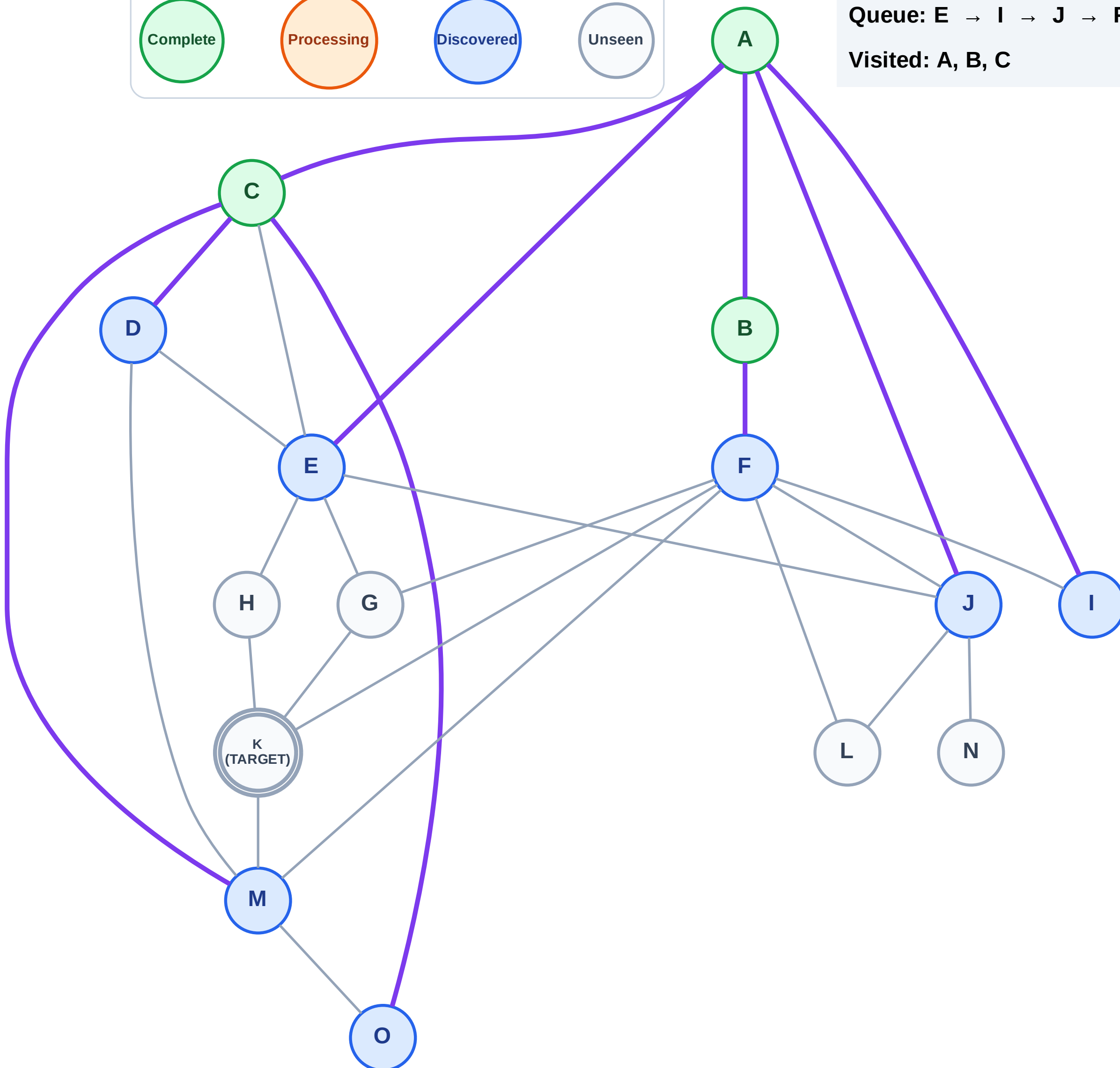
Step 7: Discover D, M, O from C

Legend



Queue: E → I → J → F → D → M → O

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

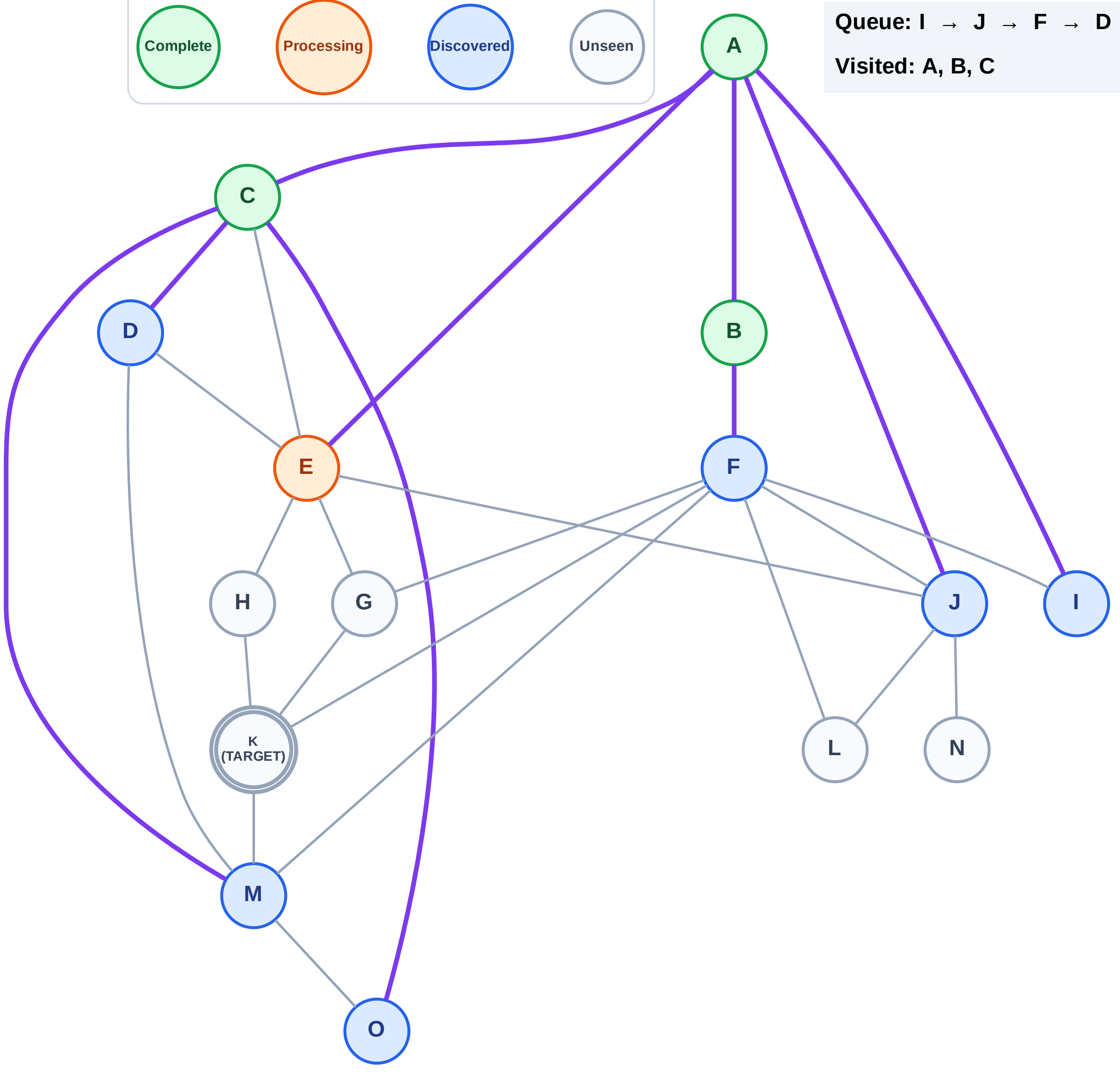
Processing

Discovered

Unseen

Queue: I → J → F → D → M → O

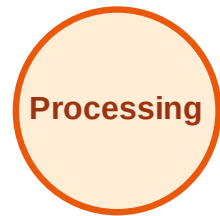
Visited: A, B, C



BFS Traversal

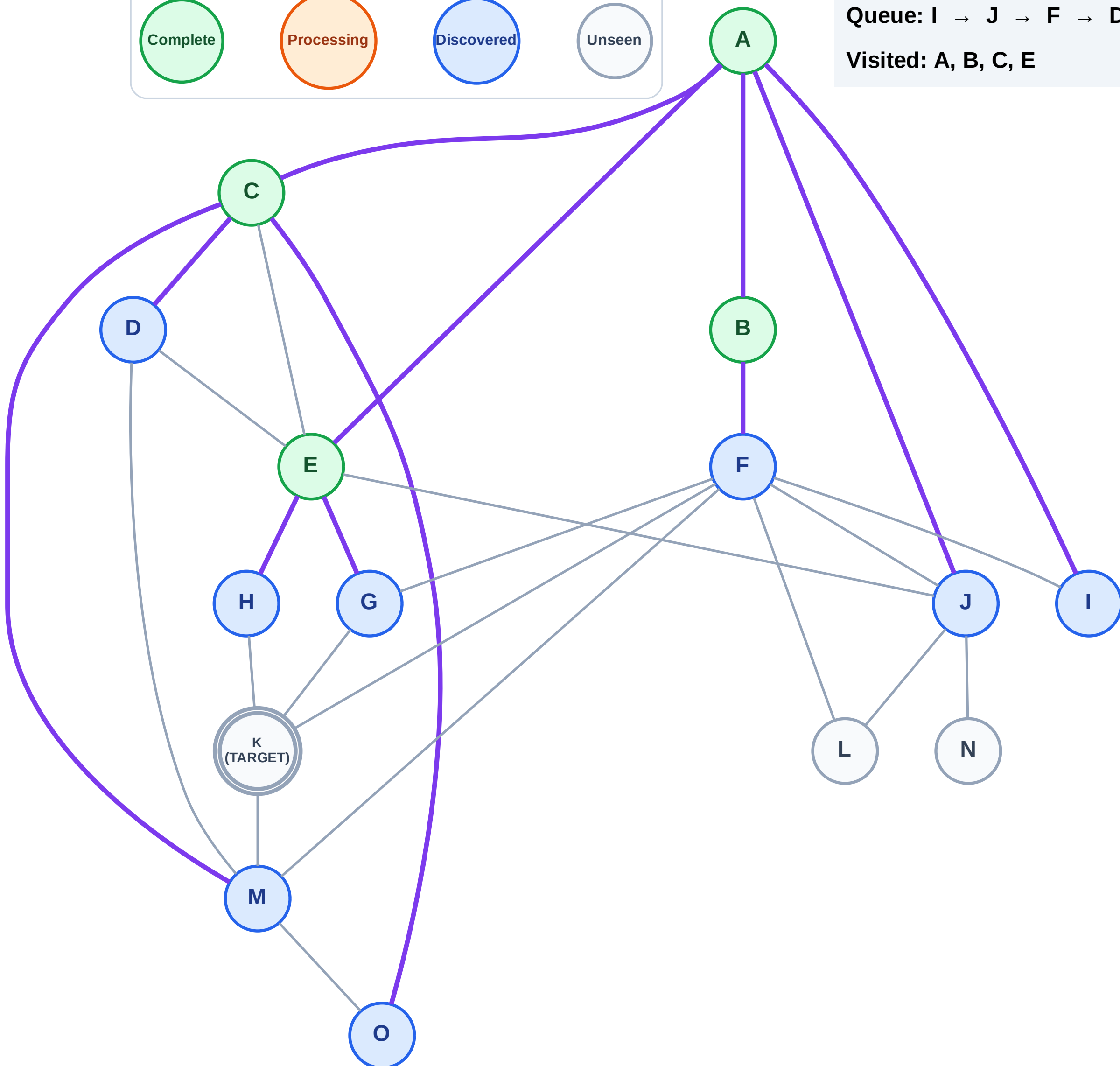
Step 9: Discover G, H from E

Legend



Queue: I → J → F → D → M → O → G → H

Visited: A, B, C, E



BFS Traversal

Step 10: Process node I

Legend

Complete

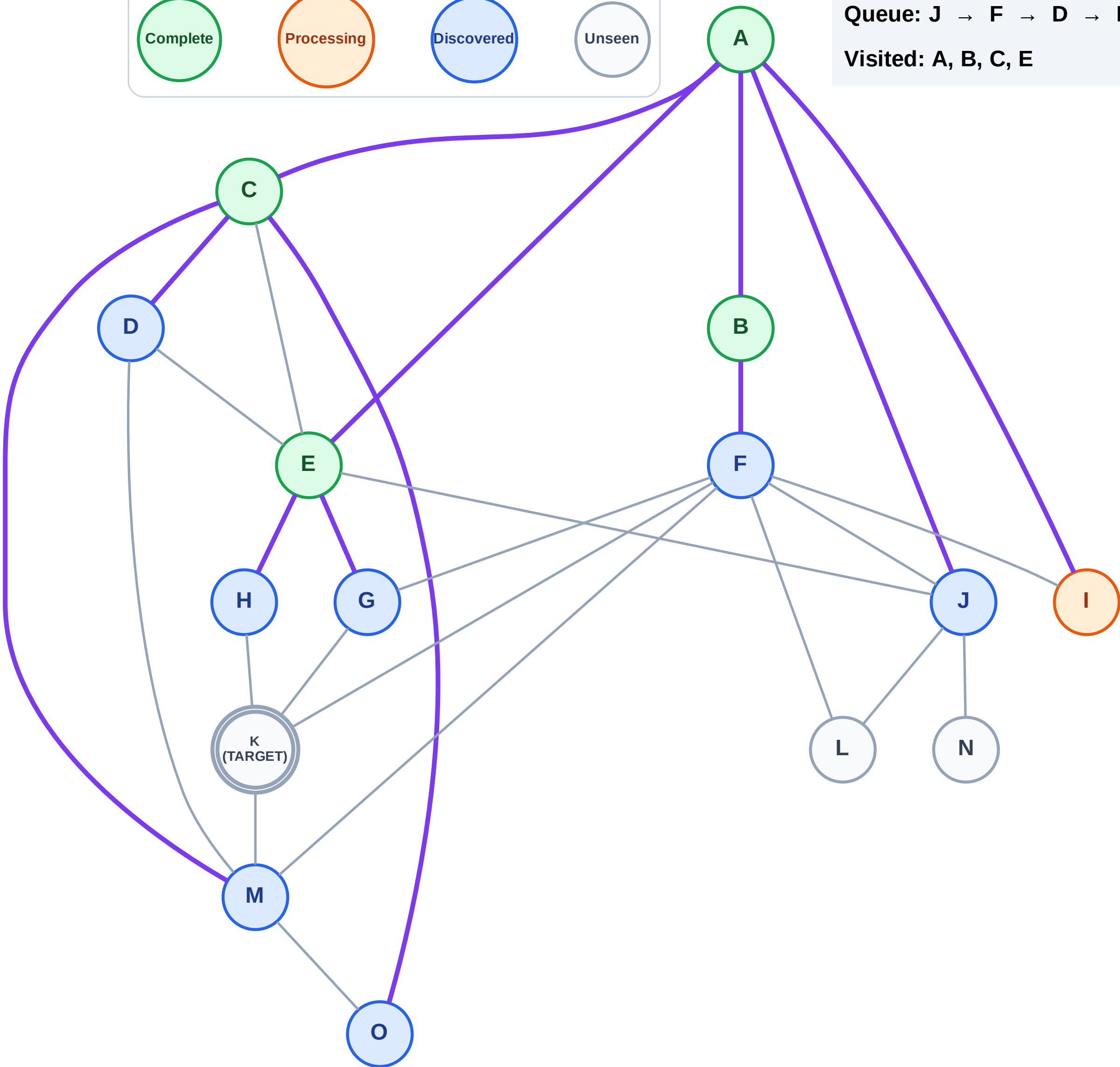
Processing

Discovered

Unseen

Queue: J → F → D → M → O → G → H

Visited: A, B, C, E



BFS Traversal

Step 11: No new neighbors from I

Legend

Complete

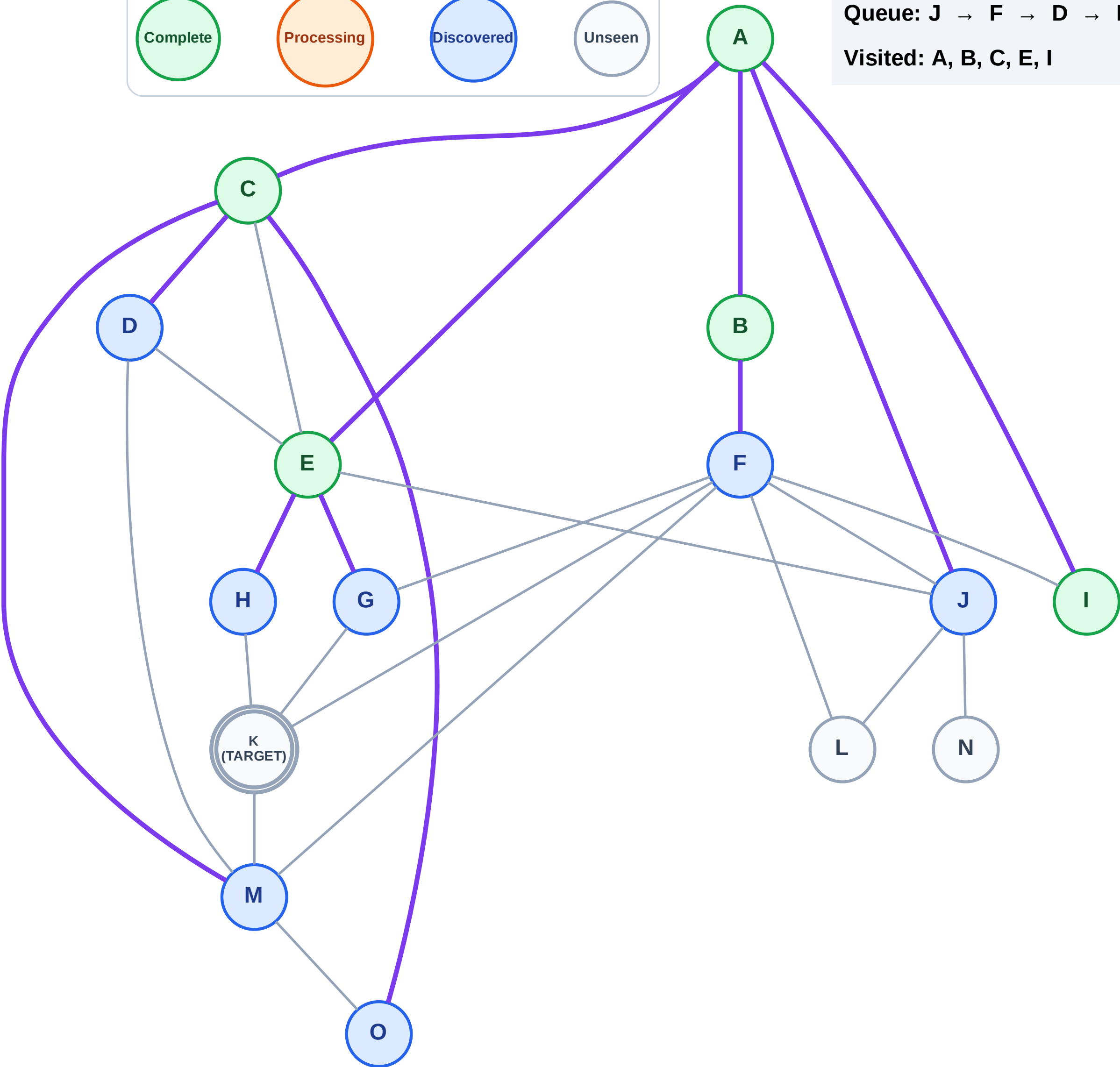
Processing

Discovered

Unseen

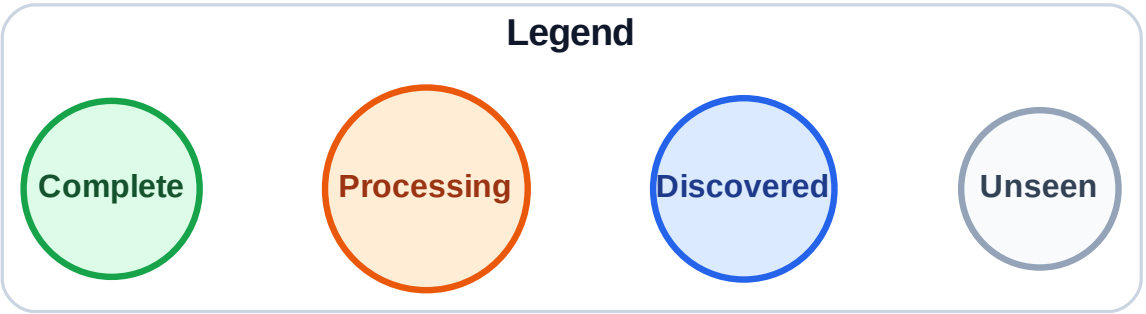
Queue: J → F → D → M → O → G → H

Visited: A, B, C, E, I



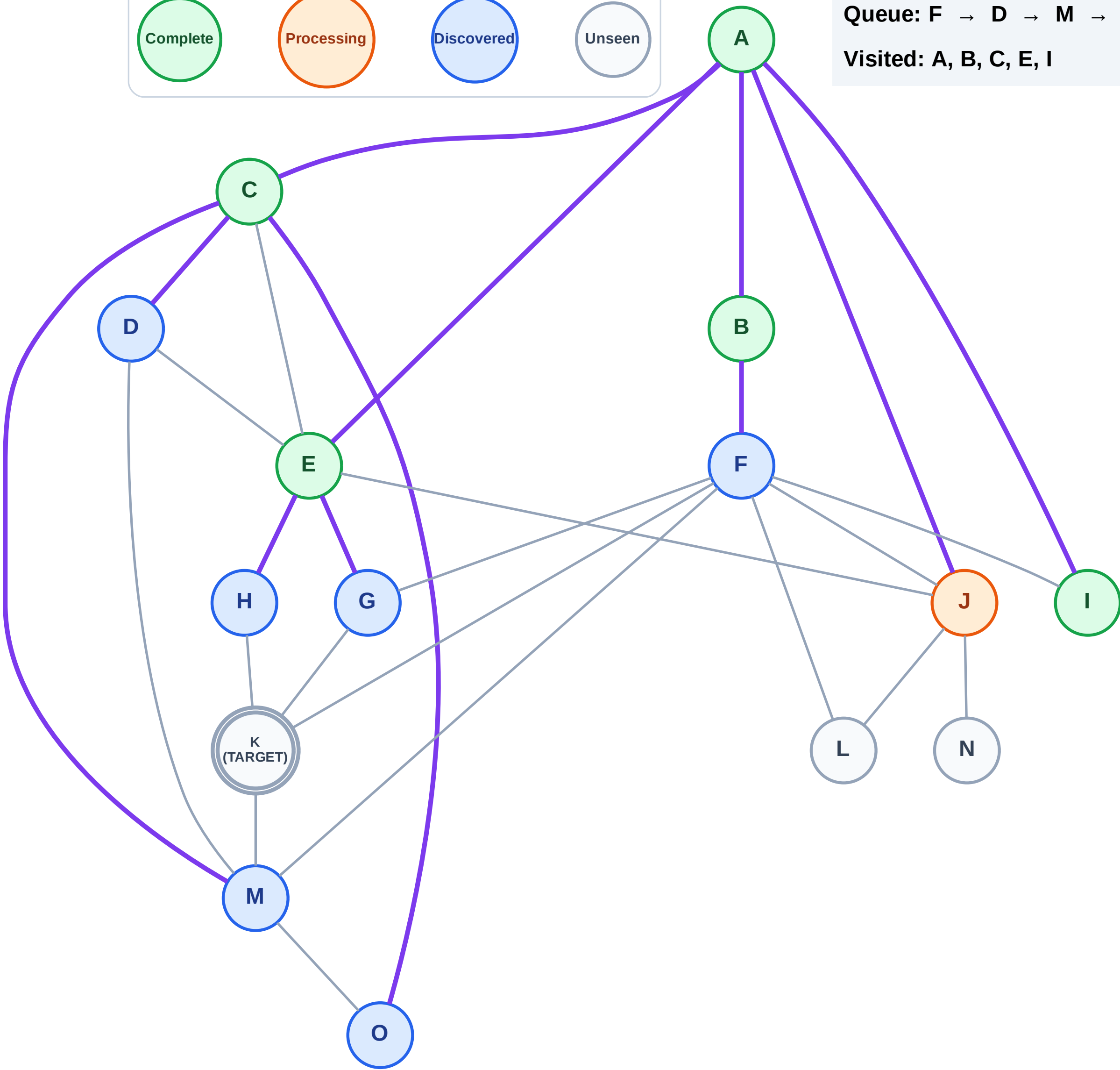
BFS Traversal

Step 12: Process node J



Queue: F → D → M → O → G → H

Visited: A, B, C, E, I



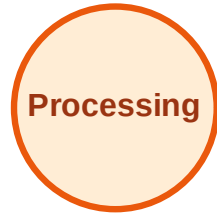
BFS Traversal

Step 13: Discover L, N from J

Legend



Complete



Processing



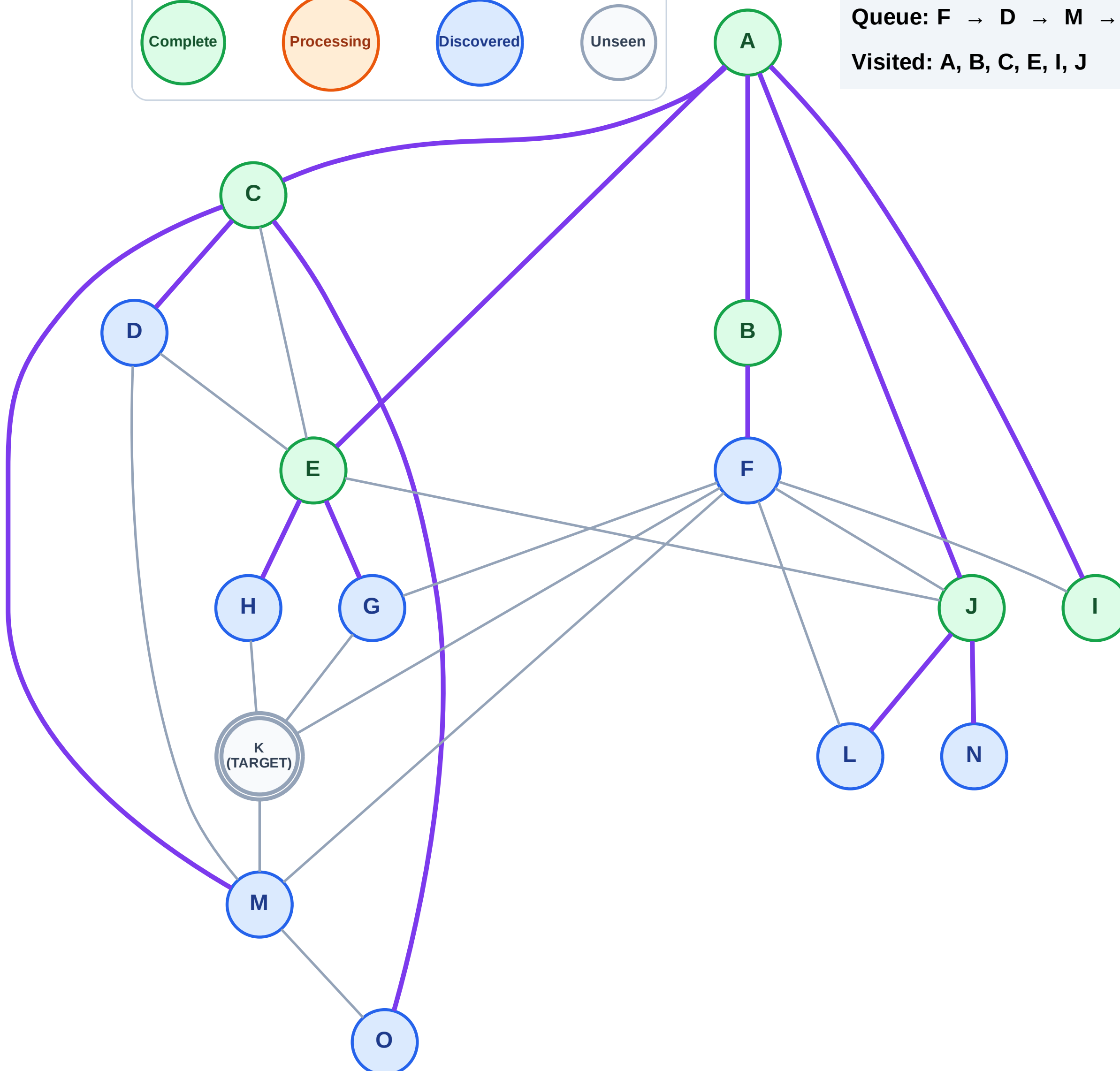
Discovered



Unseen

Queue: F → D → M → O → G → H → L → N

Visited: A, B, C, E, I, J



BFS Traversal

Step 14: Process node F

Legend

Complete

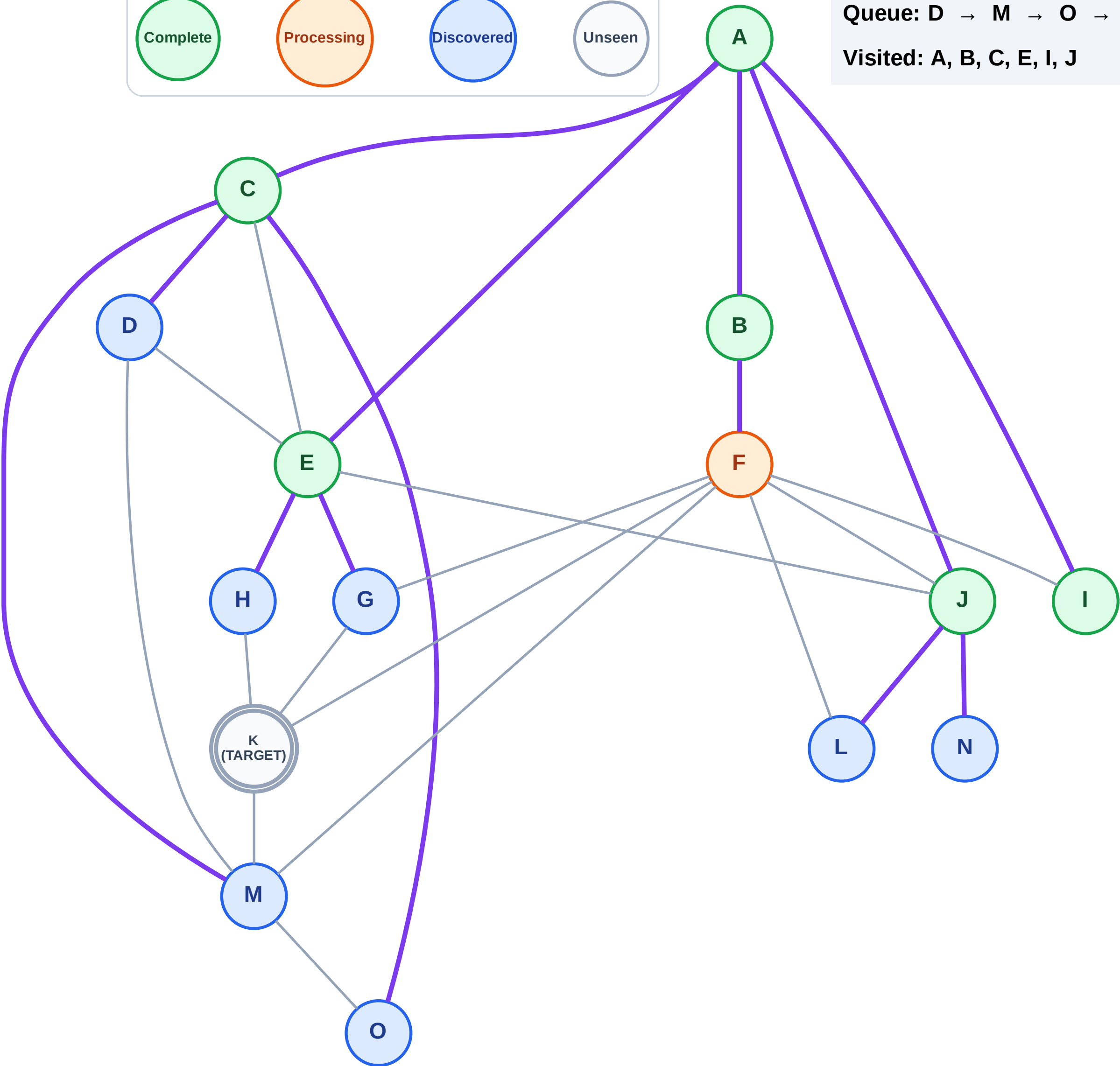
Processing

Discovered

Unseen

Queue: D → M → O → G → H → L → N

Visited: A, B, C, E, I, J



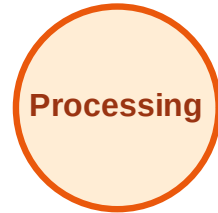
BFS Traversal

Step 15: Discover K from F

Legend



Complete



Processing



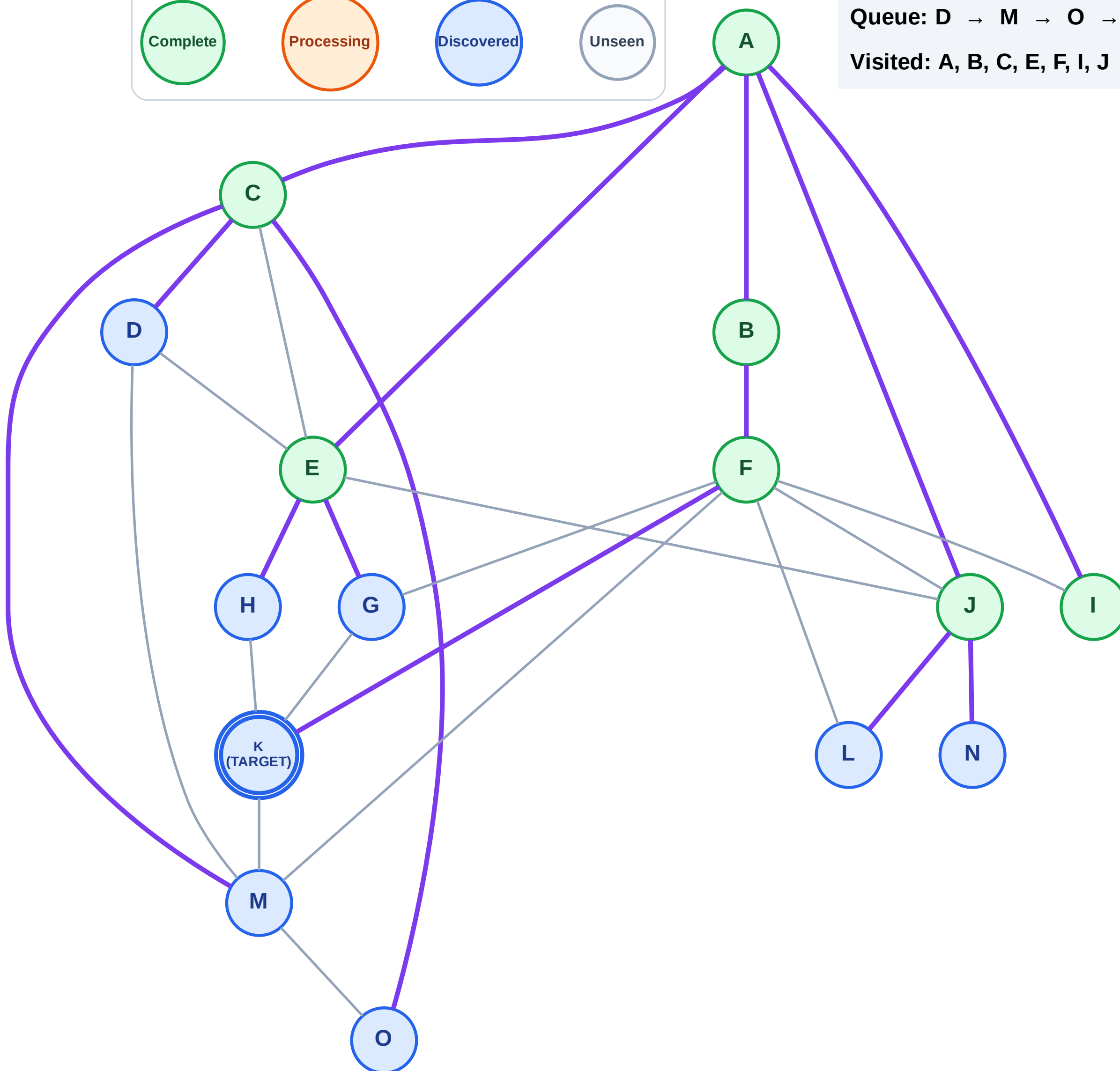
Discovered



Unseen

Queue: D → M → O → G → H → L → N → K

Visited: A, B, C, E, F, I, J



BFS Traversal

Step 16: Process node D

Legend

Complete

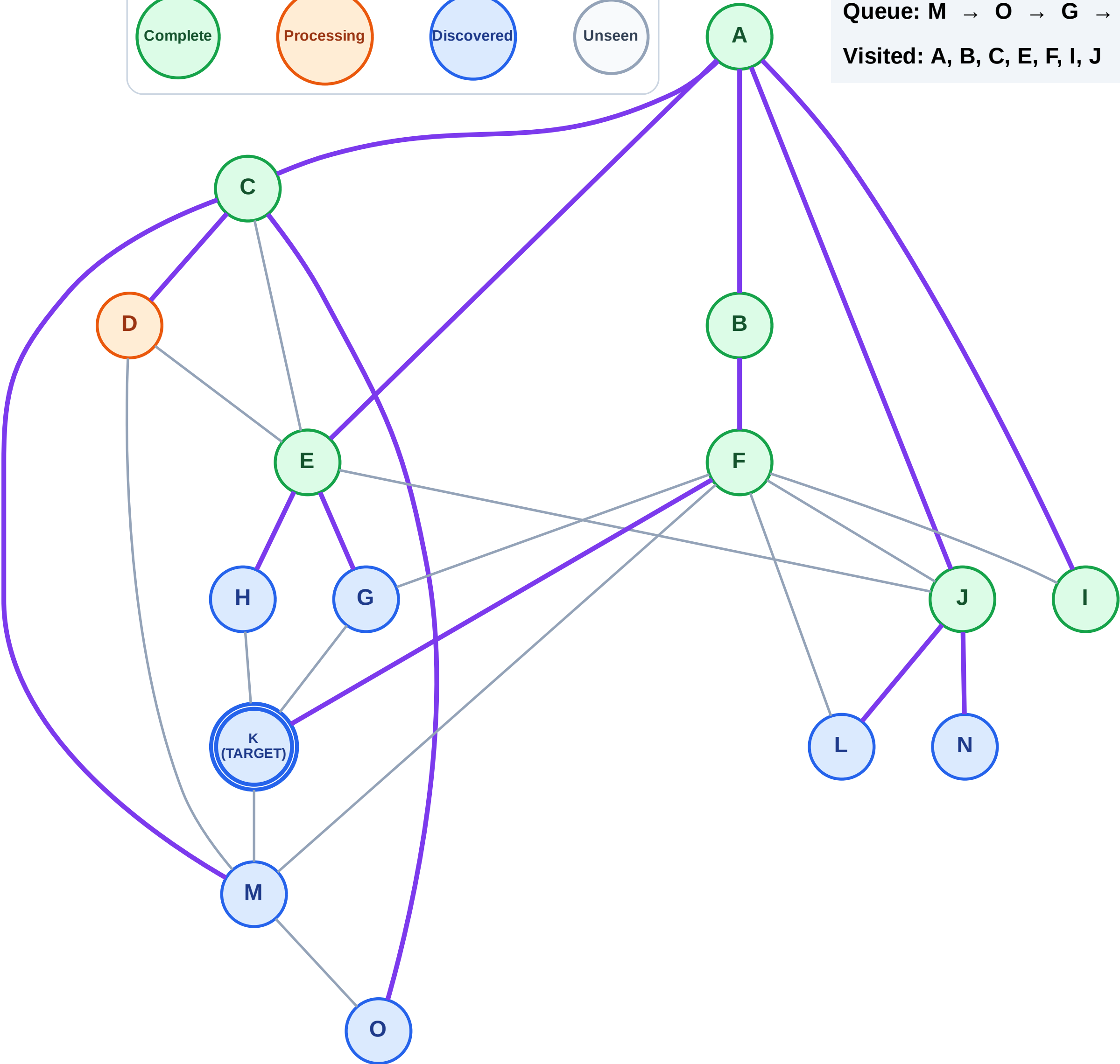
Processing

Discovered

Unseen

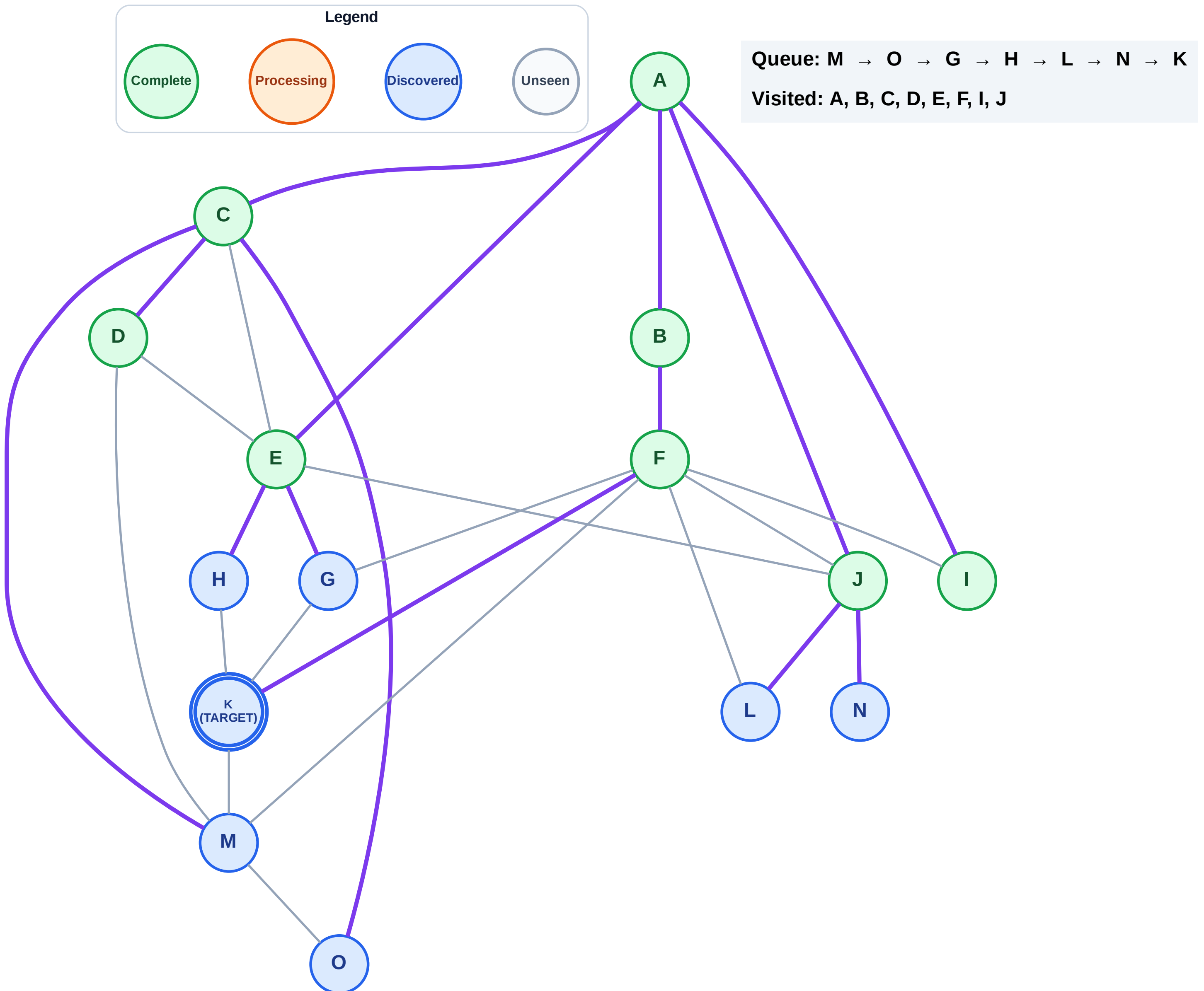
Queue: M → O → G → H → L → N → K

Visited: A, B, C, E, F, I, J



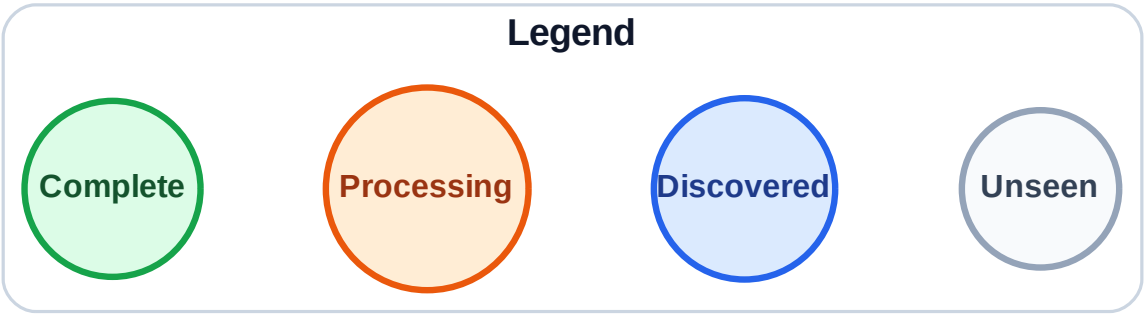
Step 17: No new neighbors from D

Step 17: No new neighbors from D

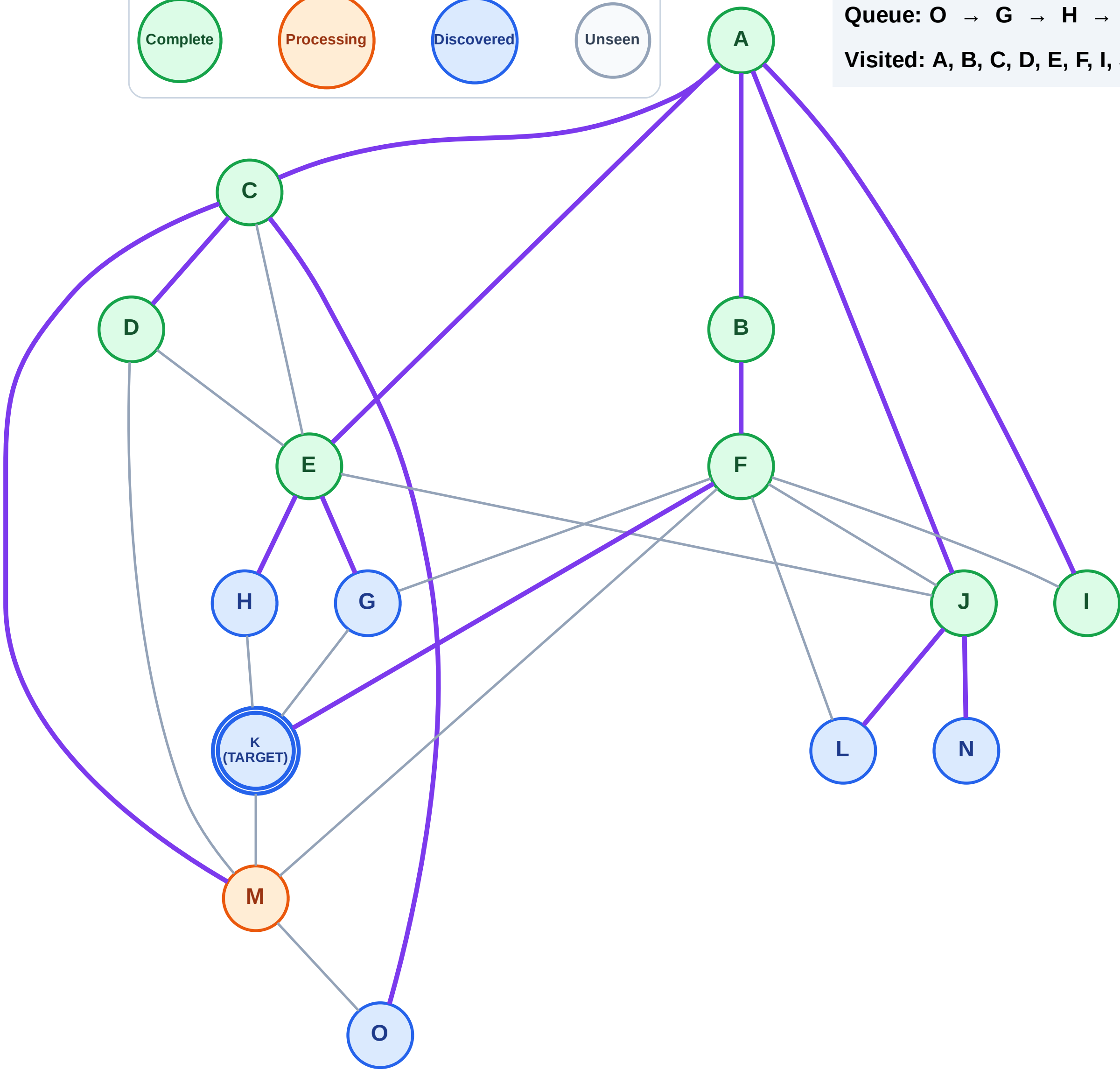


BFS Traversal

Step 18: Process node M



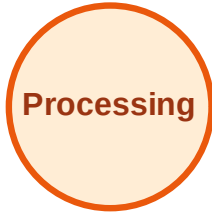
Queue: O → G → H → L → N → K
Visited: A, B, C, D, E, F, I, J



BFS Traversal

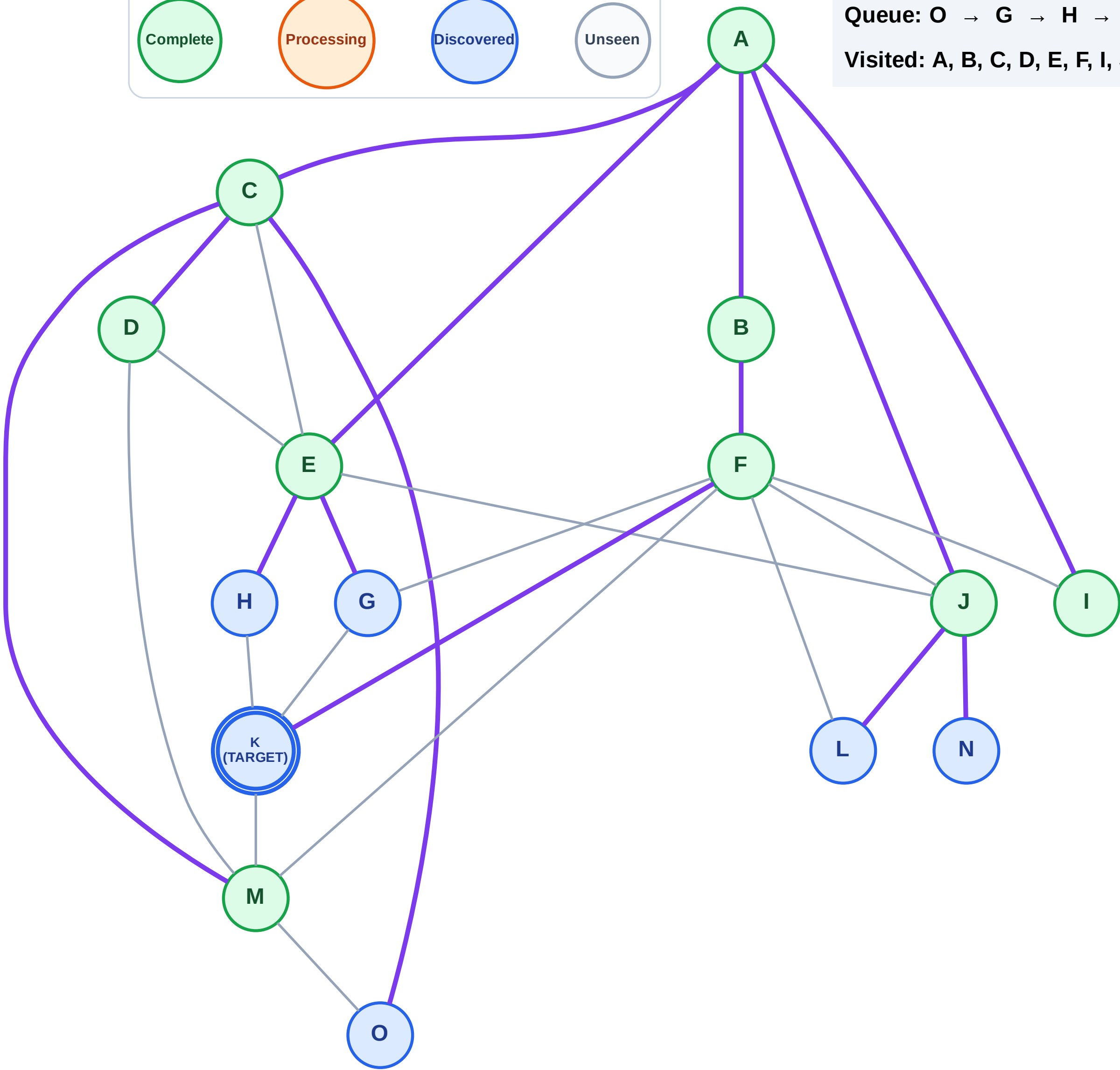
Step 19: No new neighbors from M

Legend



Queue: O → G → H → L → N → K

Visited: A, B, C, D, E, F, I, J, M



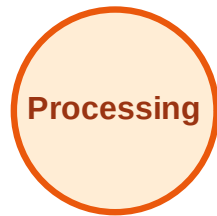
BFS Traversal

Step 20: Process node O

Legend



Complete



Processing



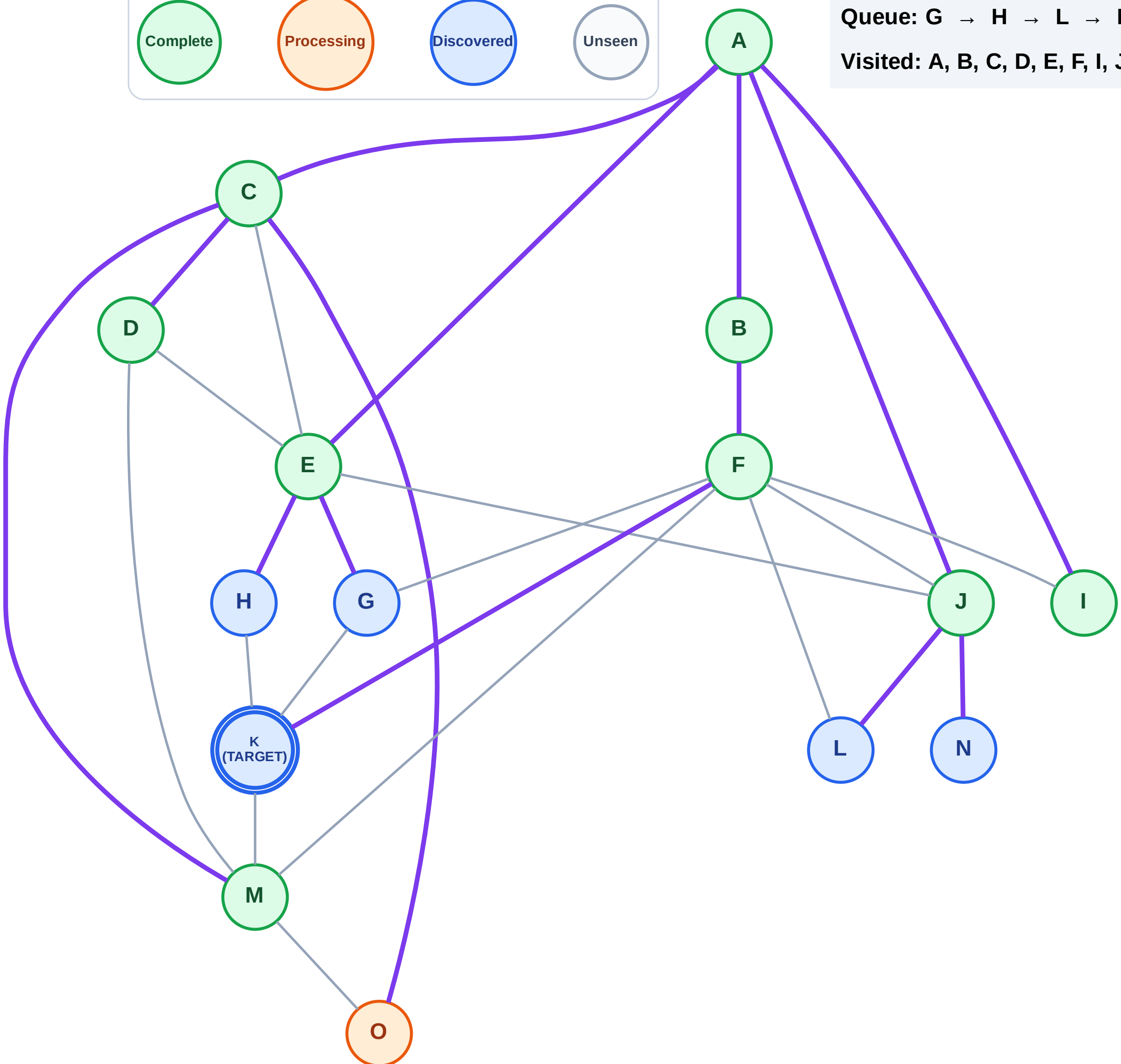
Discovered



Unseen

Queue: G → H → L → N → K

Visited: A, B, C, D, E, F, I, J, M



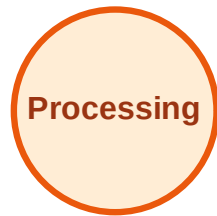
BFS Traversal

Step 21: No new neighbors from O

Legend



Complete



Processing



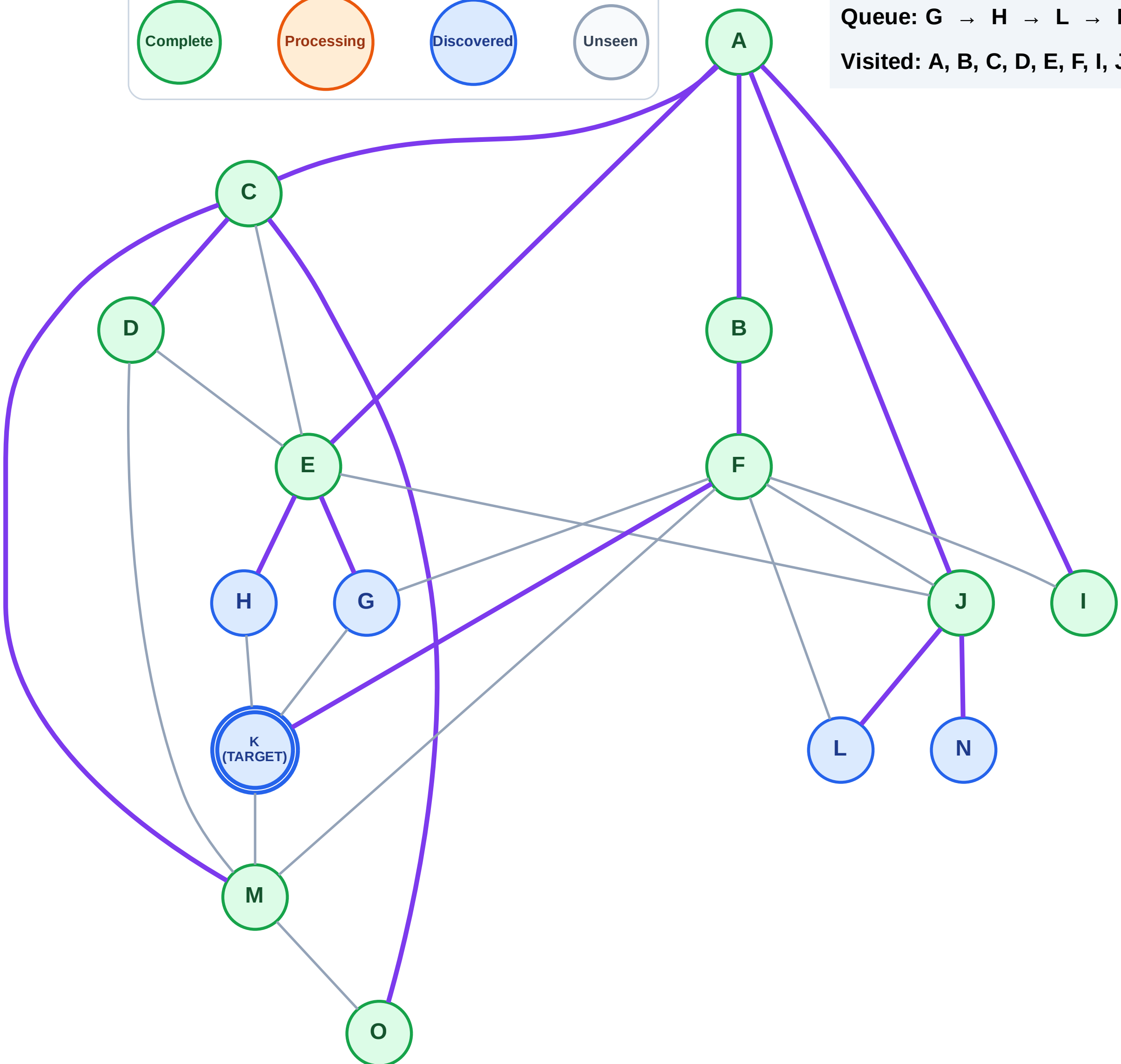
Discovered



Unseen

Queue: G → H → L → N → K

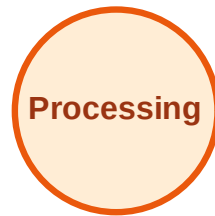
Visited: A, B, C, D, E, F, I, J, M, O



BFS Traversal

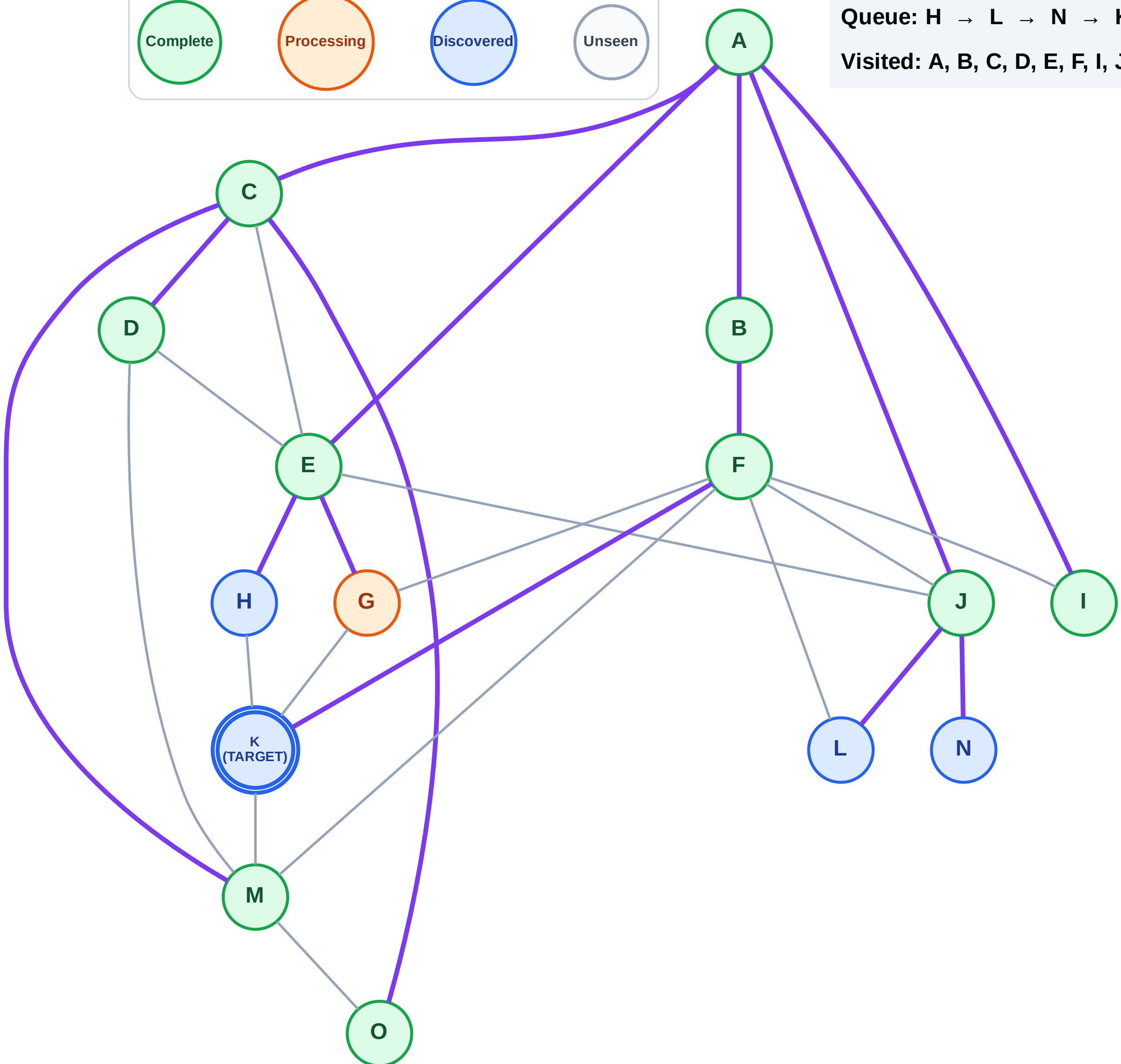
Step 22: Process node G

Legend



Queue: H → L → N → K

Visited: A, B, C, D, E, F, I, J, M, O



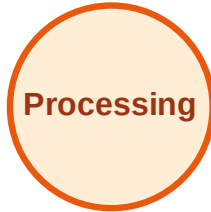
BFS Traversal

Step 23: No new neighbors from G

Legend



Complete



Processing



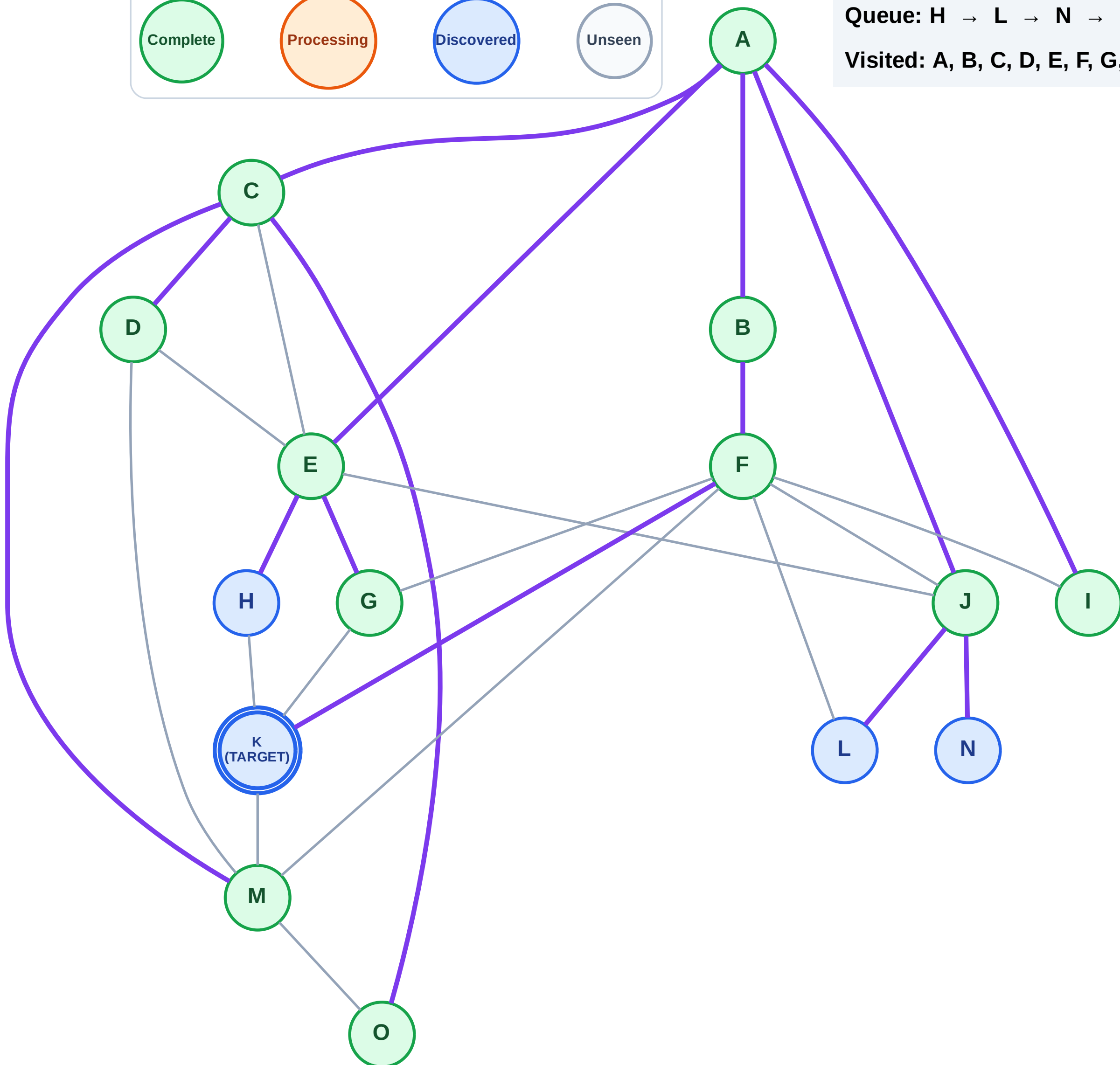
Discovered



Unseen

Queue: H → L → N → K

Visited: A, B, C, D, E, F, G, I, J, M, O



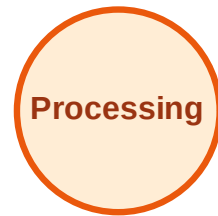
BFS Traversal

Step 24: Process node H

Legend



Complete



Processing



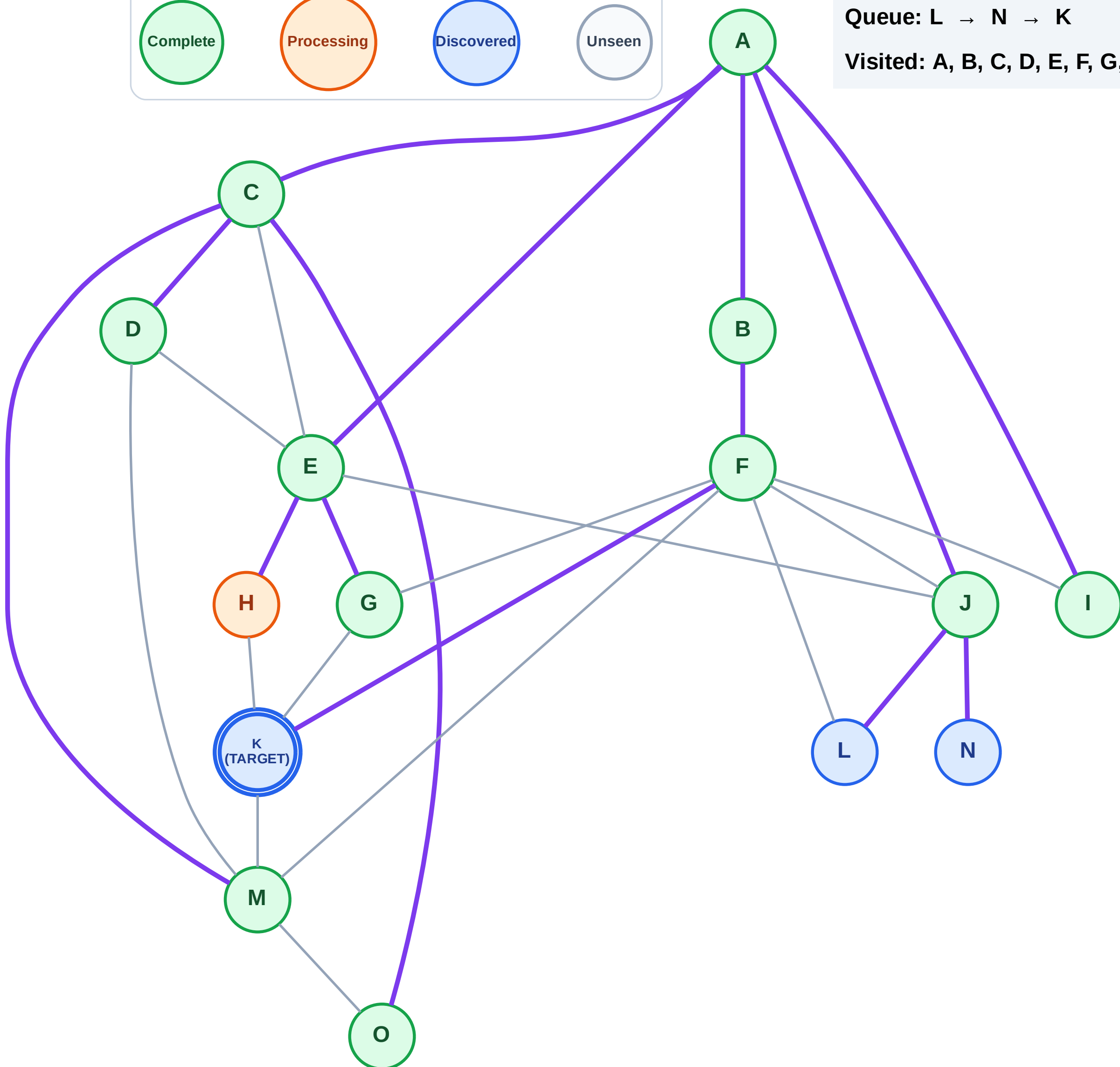
Discovered



Unseen

Queue: L → N → K

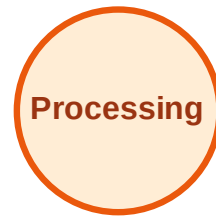
Visited: A, B, C, D, E, F, G, I, J, M, O



BFS Traversal

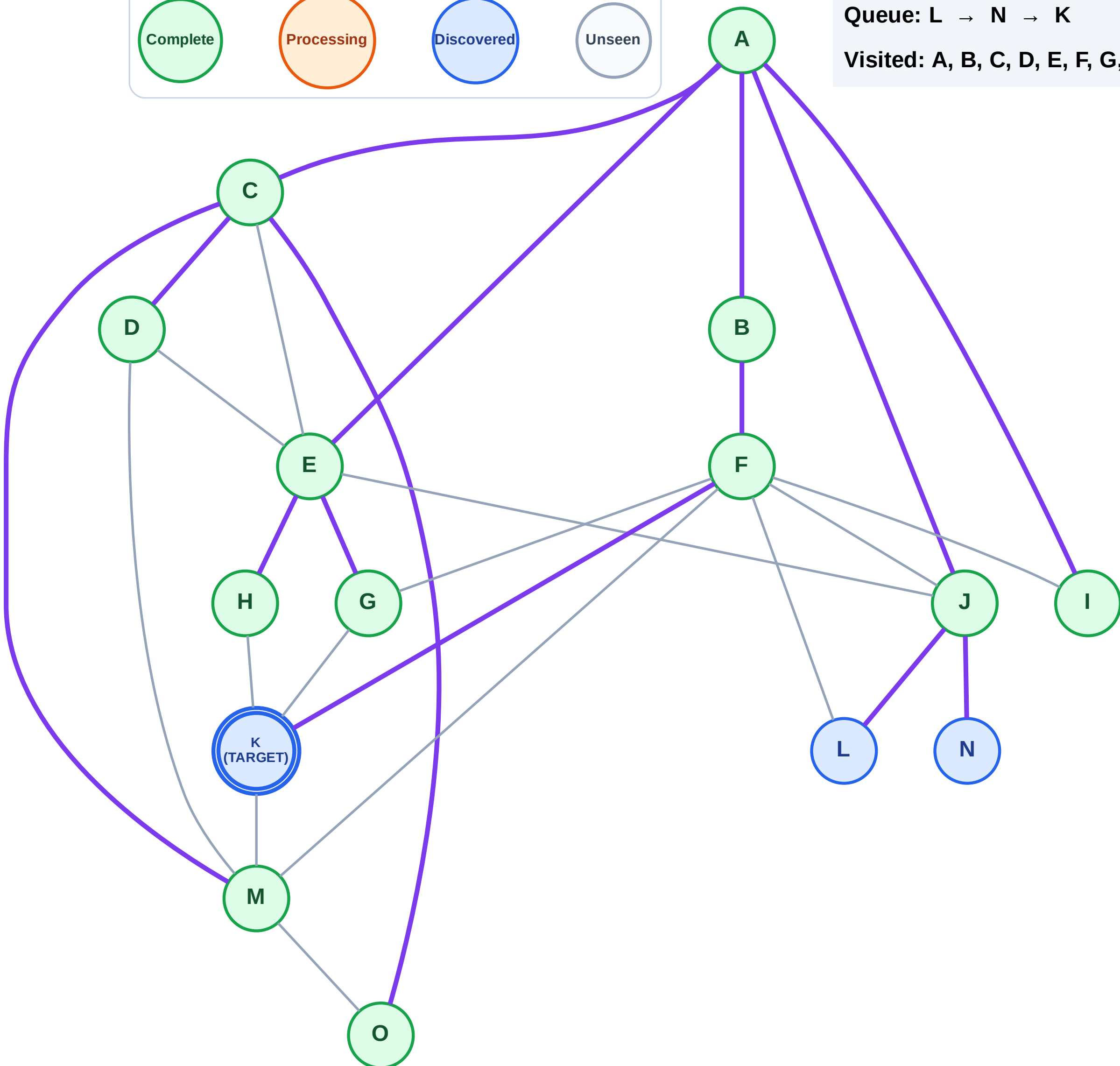
Step 25: No new neighbors from H

Legend



Queue: L → N → K

Visited: A, B, C, D, E, F, G, H, I, J, M, O



BFS Traversal

Step 26: Process node L

Legend

Complete

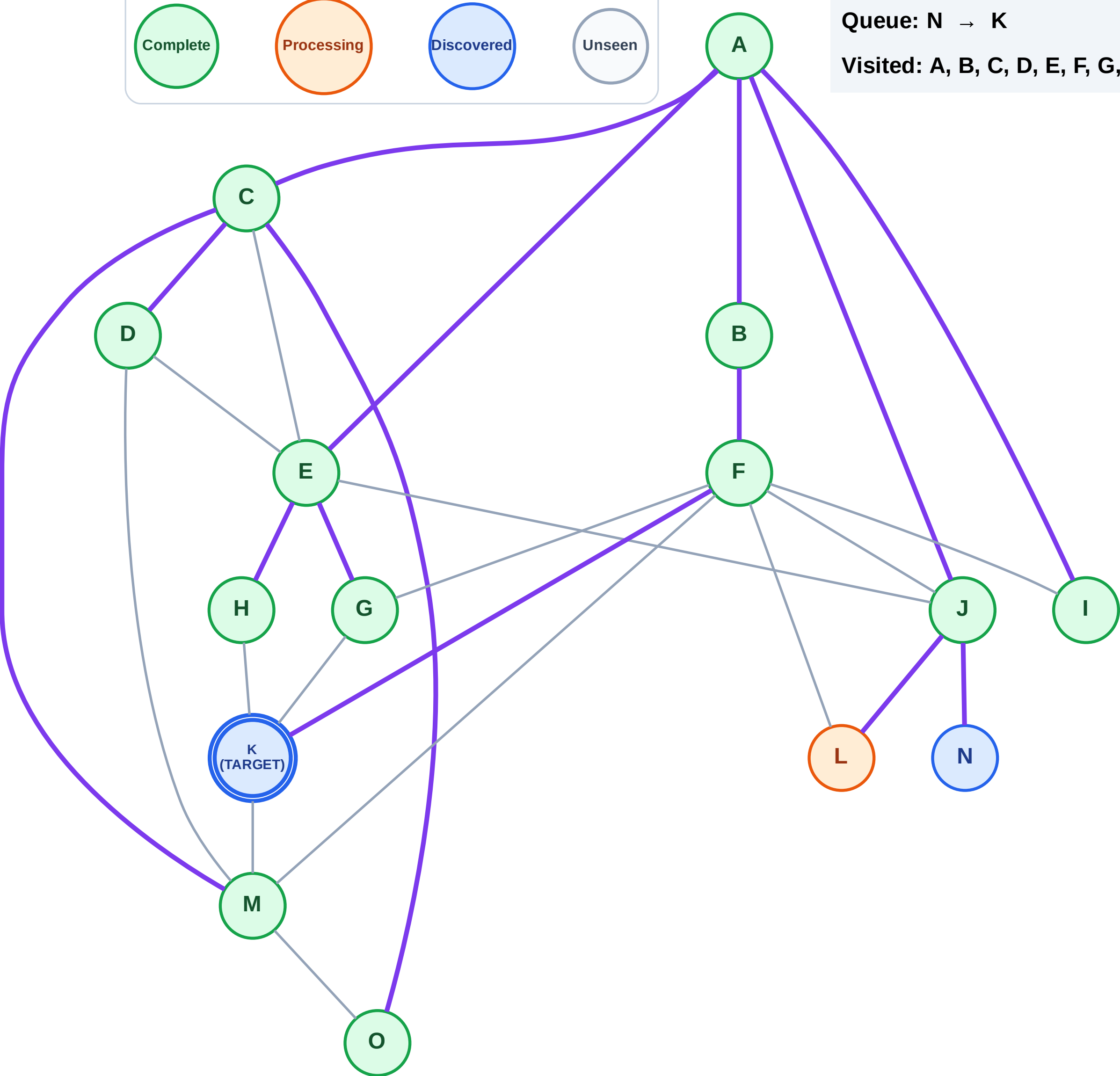
Processing

Discovered

Unseen

Queue: N → K

Visited: A, B, C, D, E, F, G, H, I, J, M, O



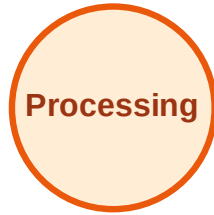
BFS Traversal

Step 27: No new neighbors from L

Legend



Complete



Processing



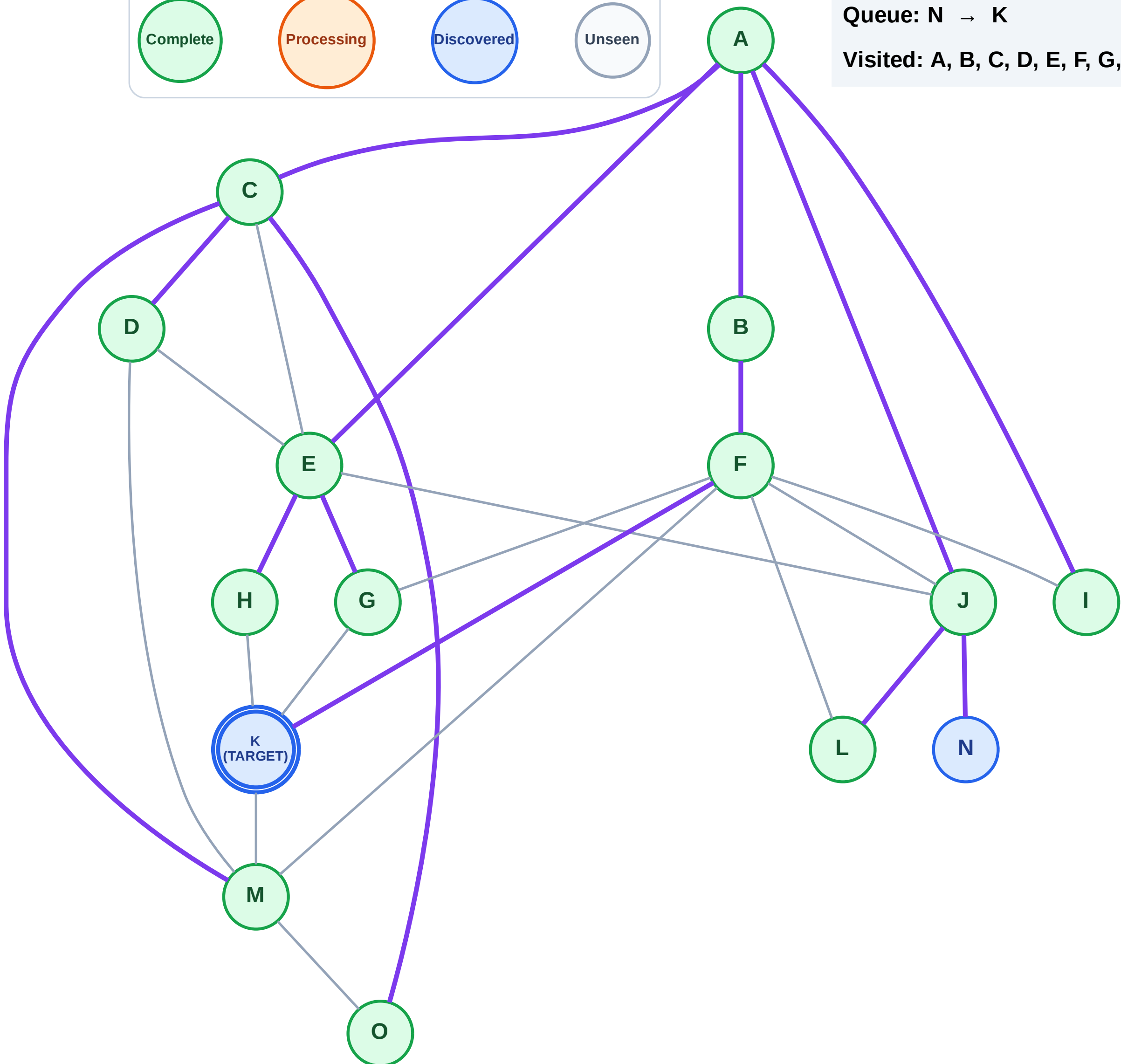
Discovered



Unseen

Queue: N → K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

Step 28: Process node N

Legend

Complete

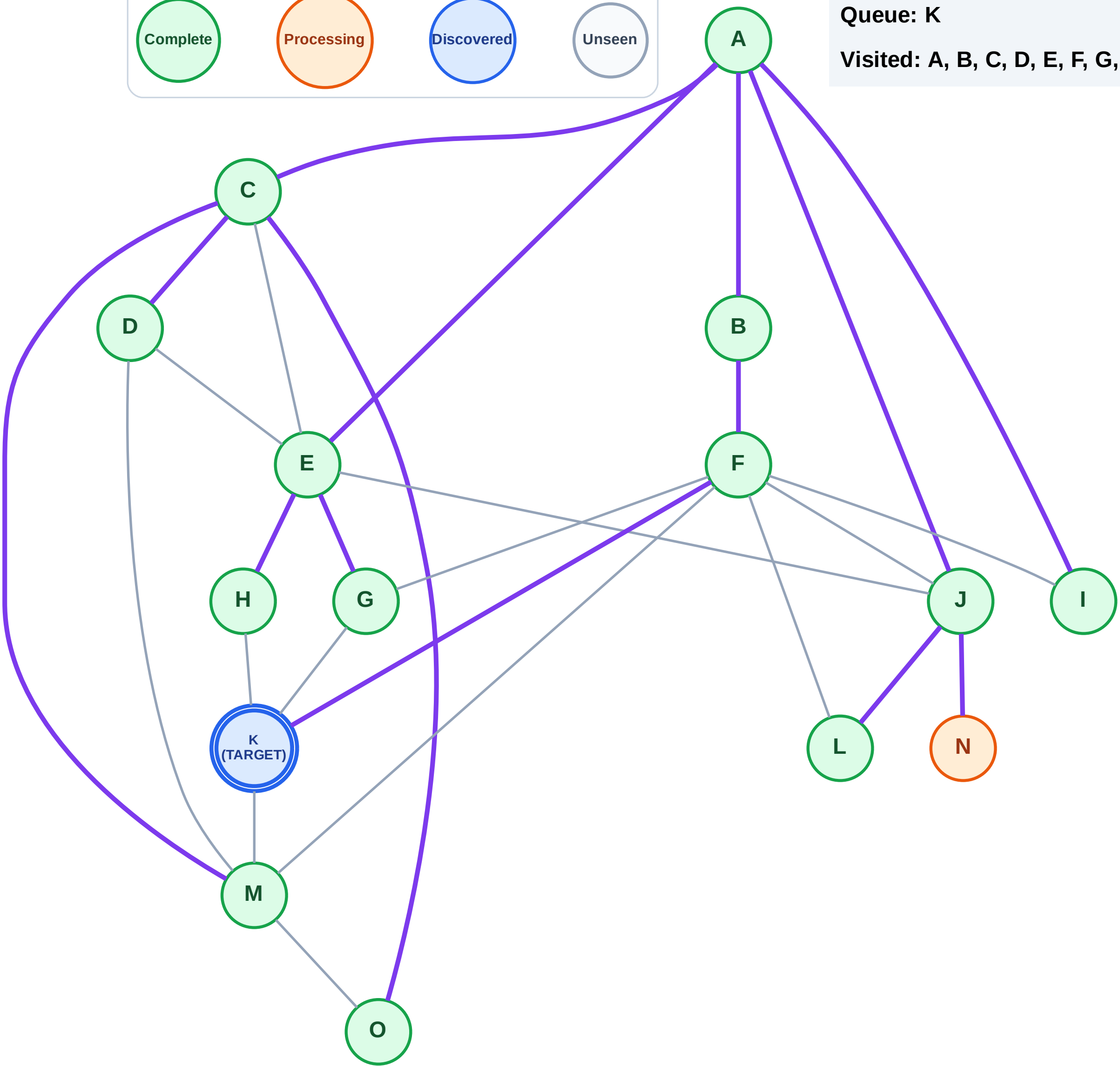
Processing

Discovered

Unseen

Queue: K

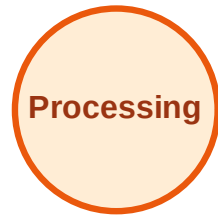
Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

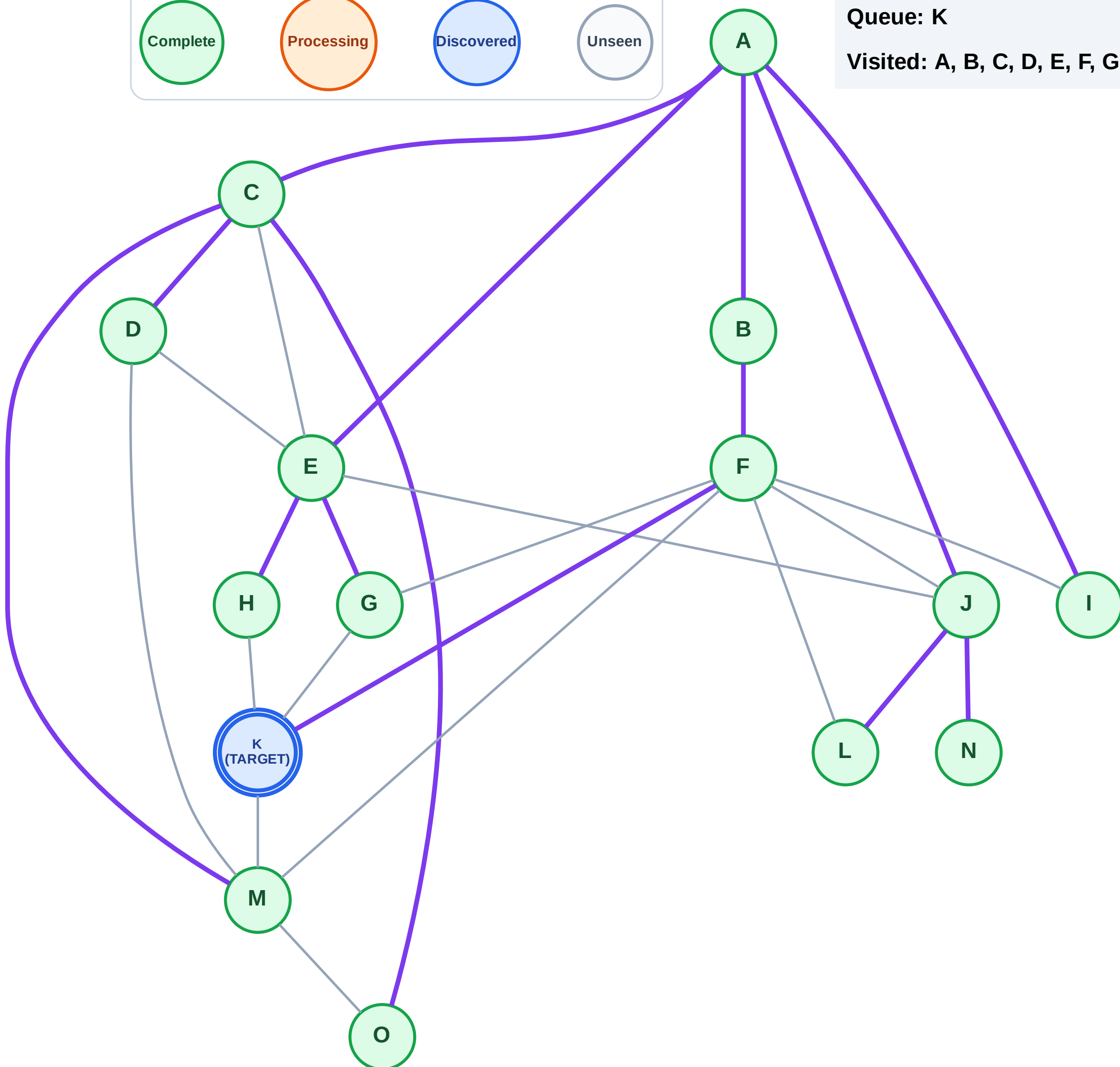
Step 29: No new neighbors from N

Legend



Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 30: Process node K

Legend

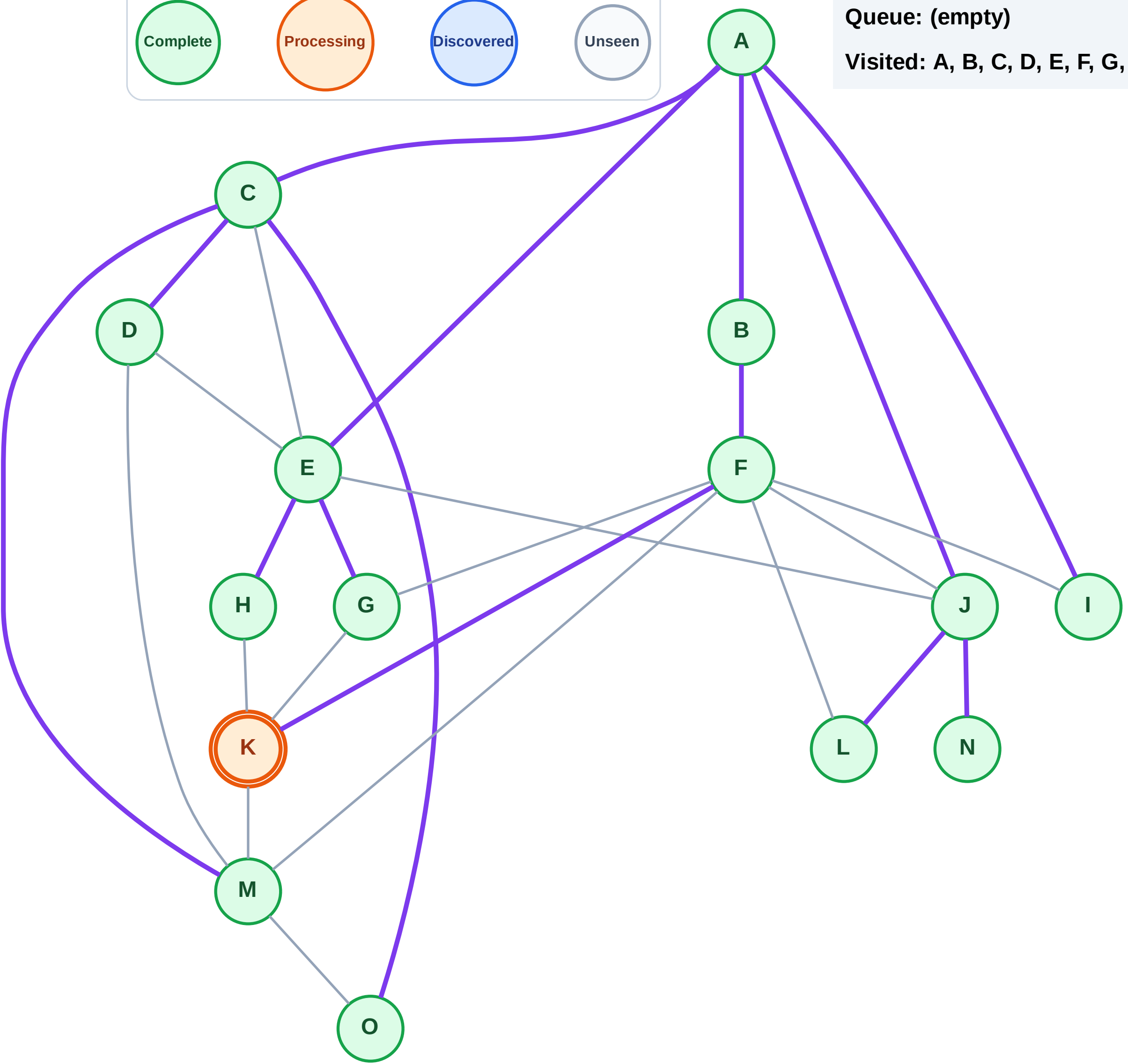
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 31: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend

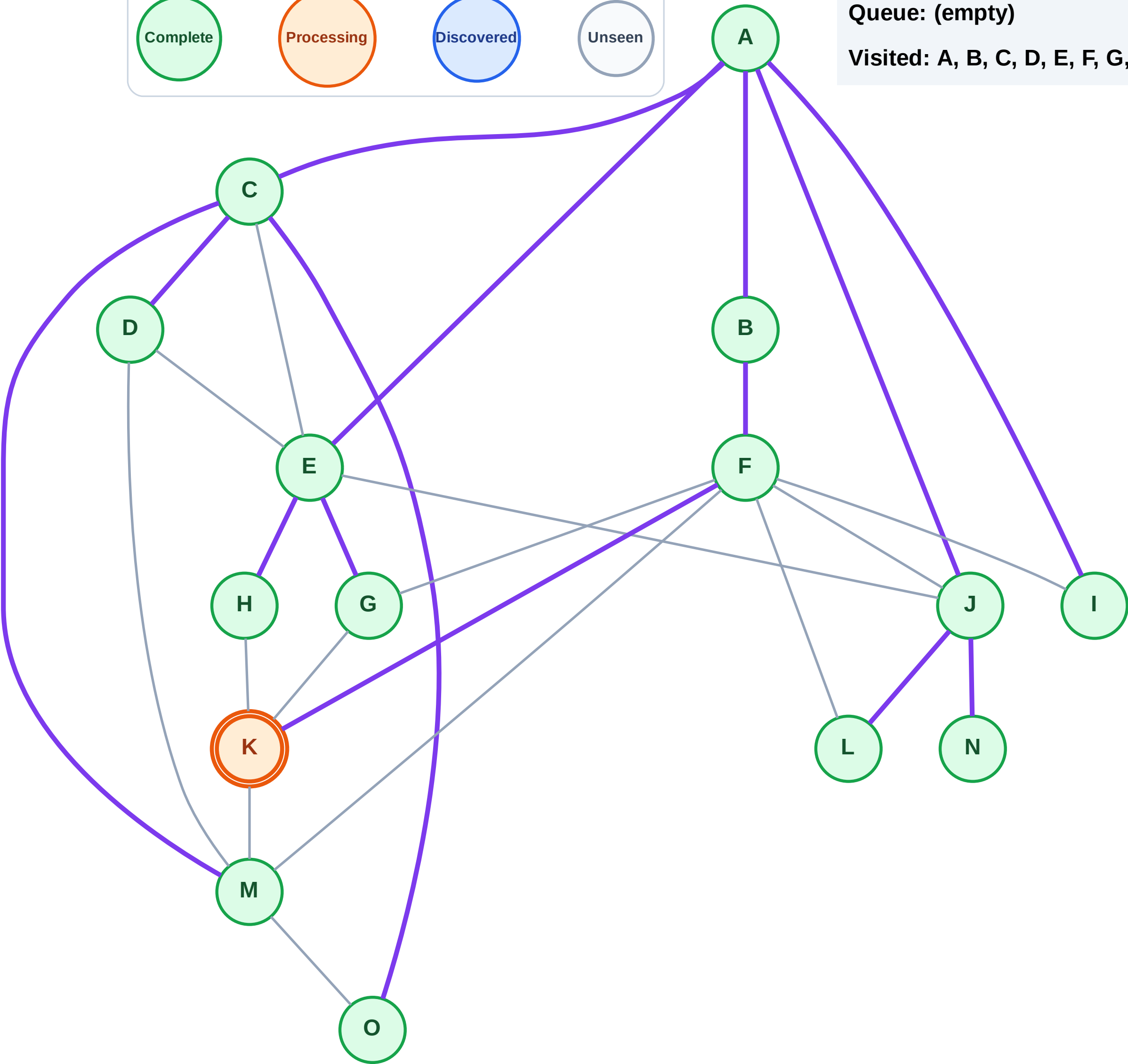
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

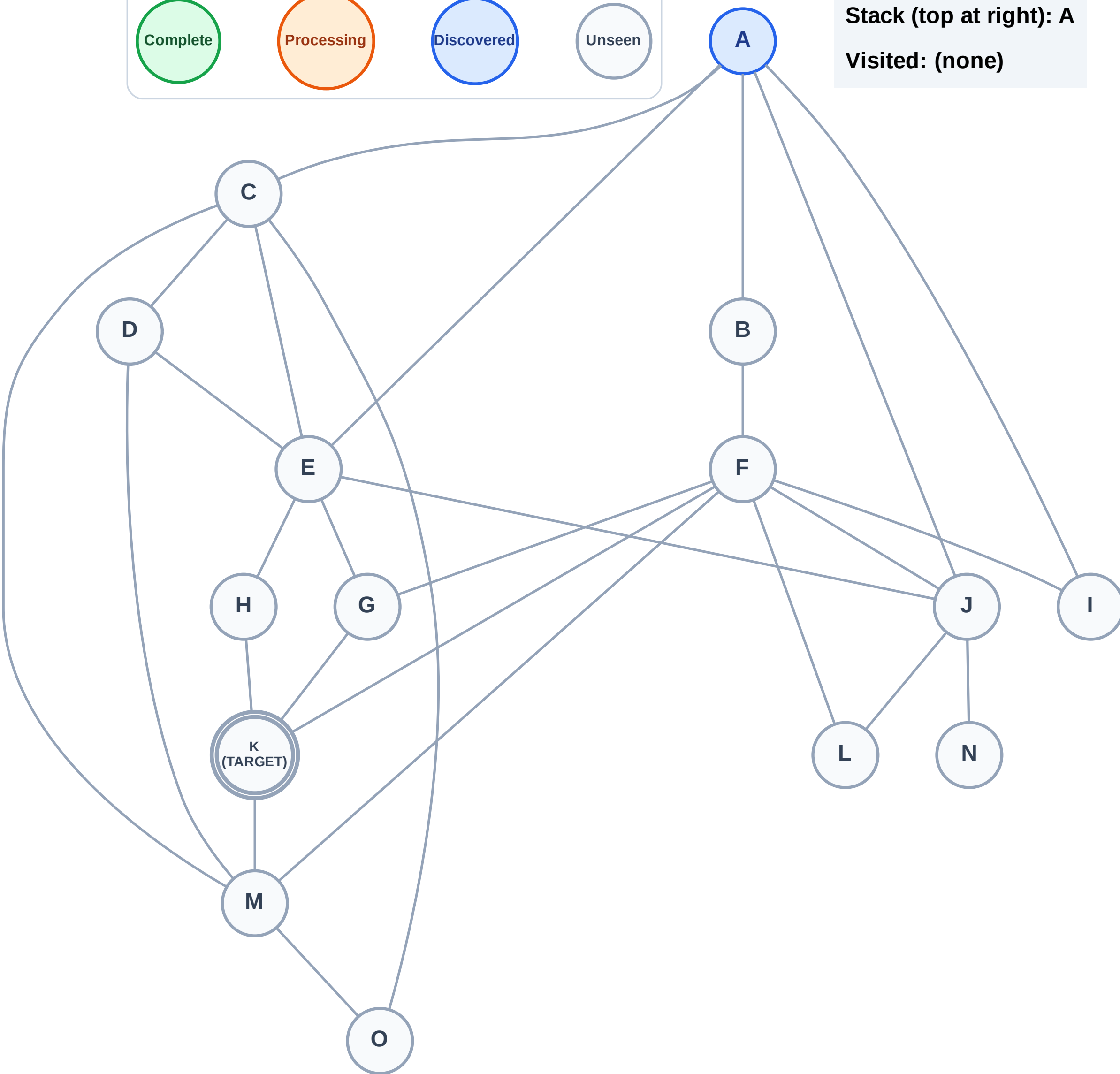
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

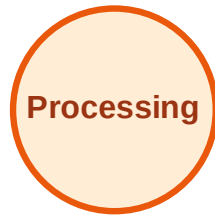
Visited: (none)



DFS Traversal

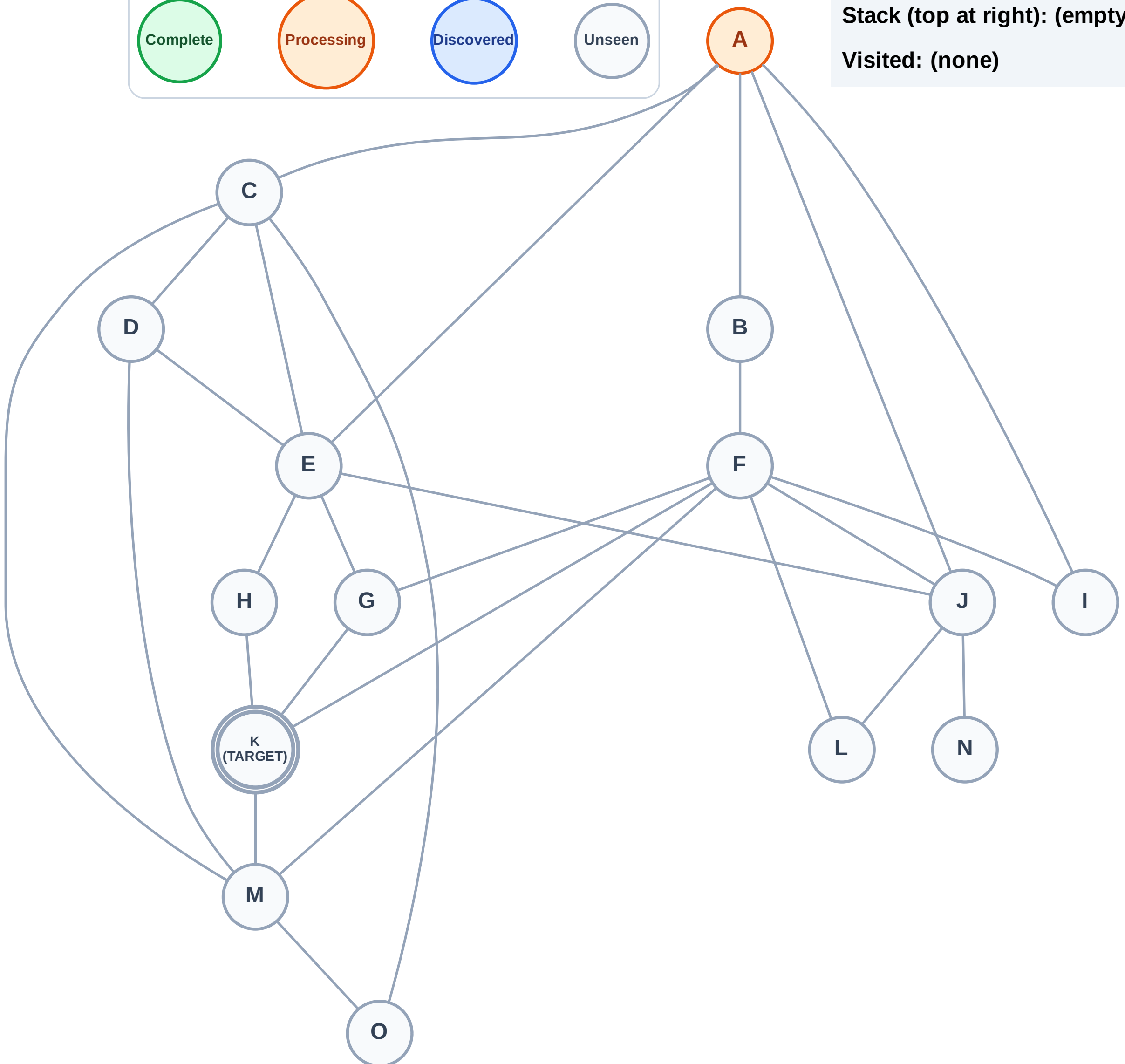
Step 2: Process node A

Legend



Stack (top at right): (empty)

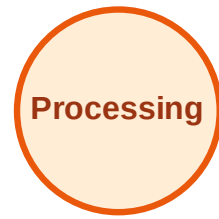
Visited: (none)



DFS Traversal

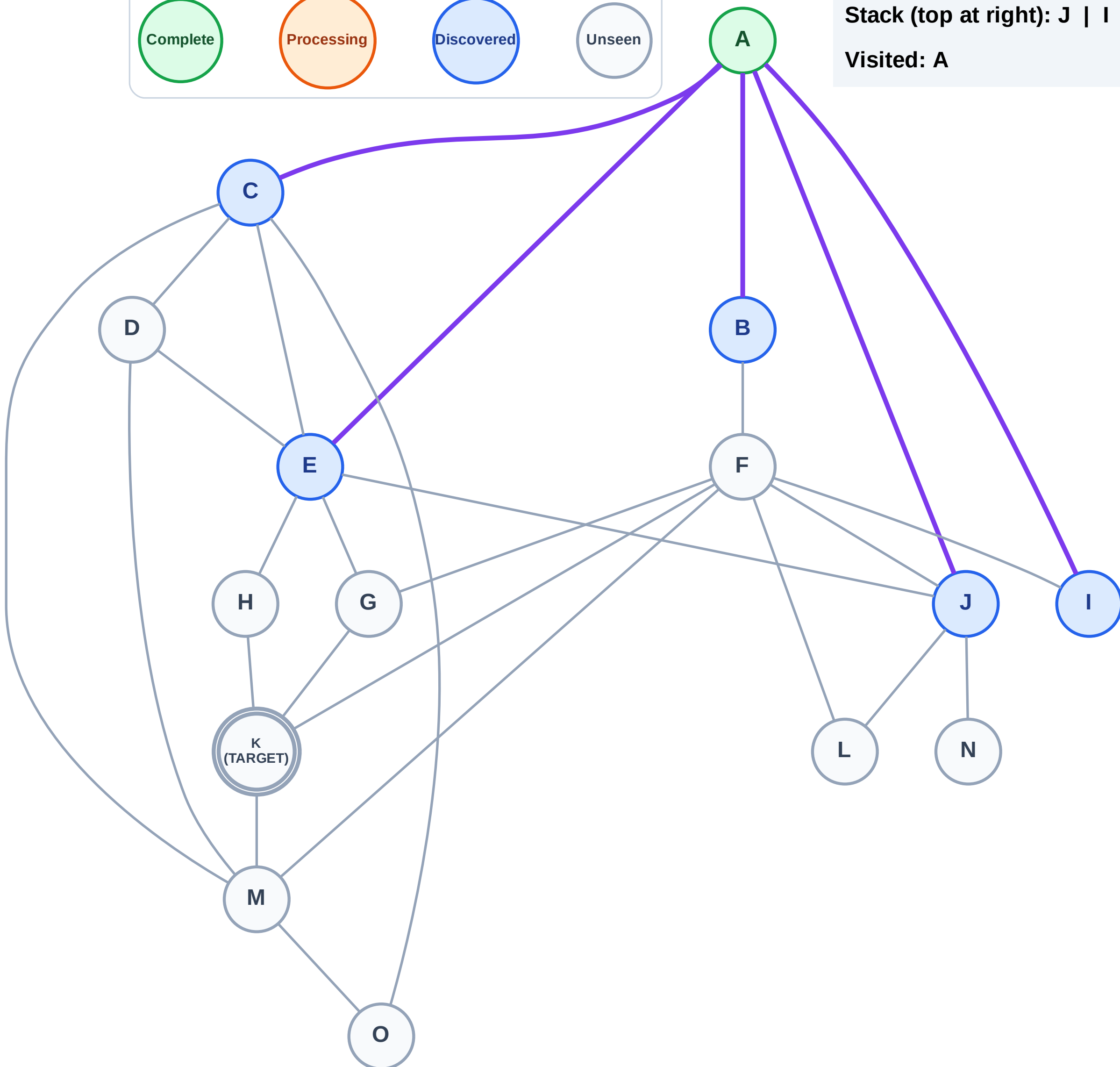
Step 3: Discover J, I, E, C, B from A

Legend



Stack (top at right): J | I | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

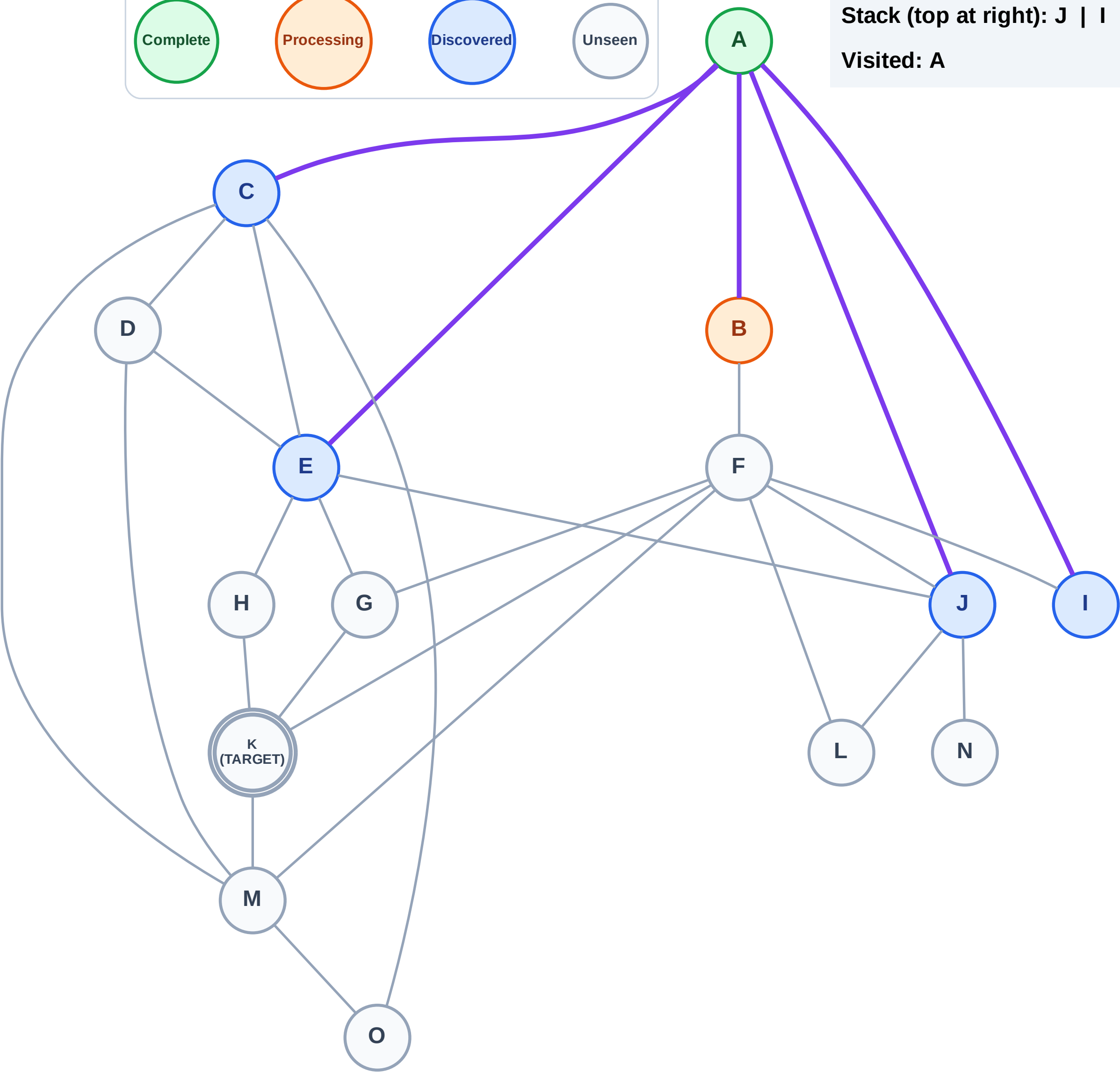
Processing

Discovered

Unseen

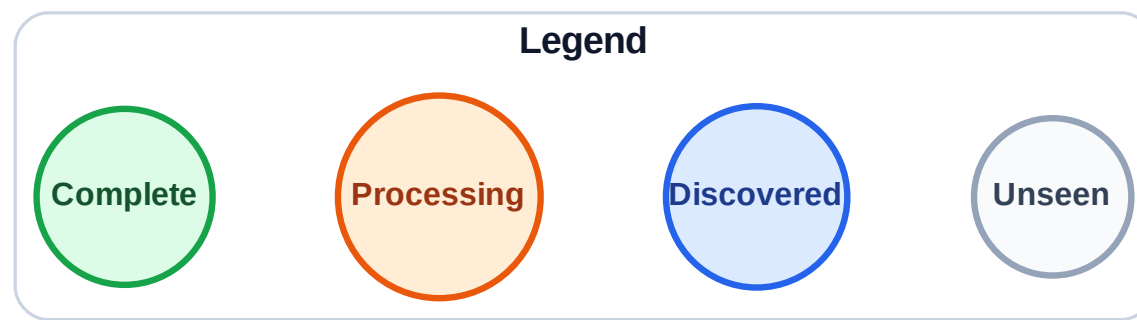
Stack (top at right): J | I | E | C

Visited: A



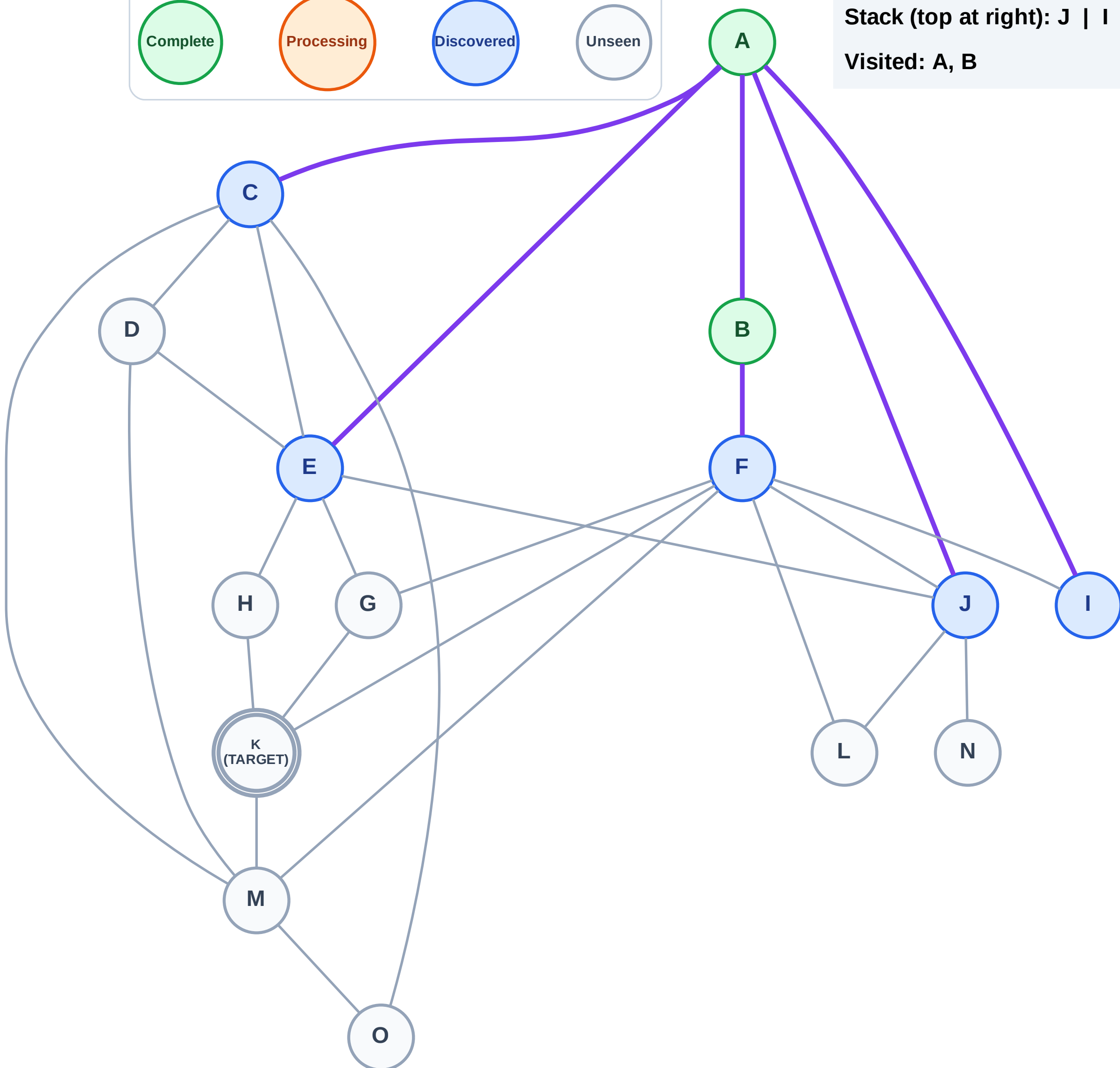
DFS Traversal

Step 5: Discover F from B



Stack (top at right): J | I | E | C | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

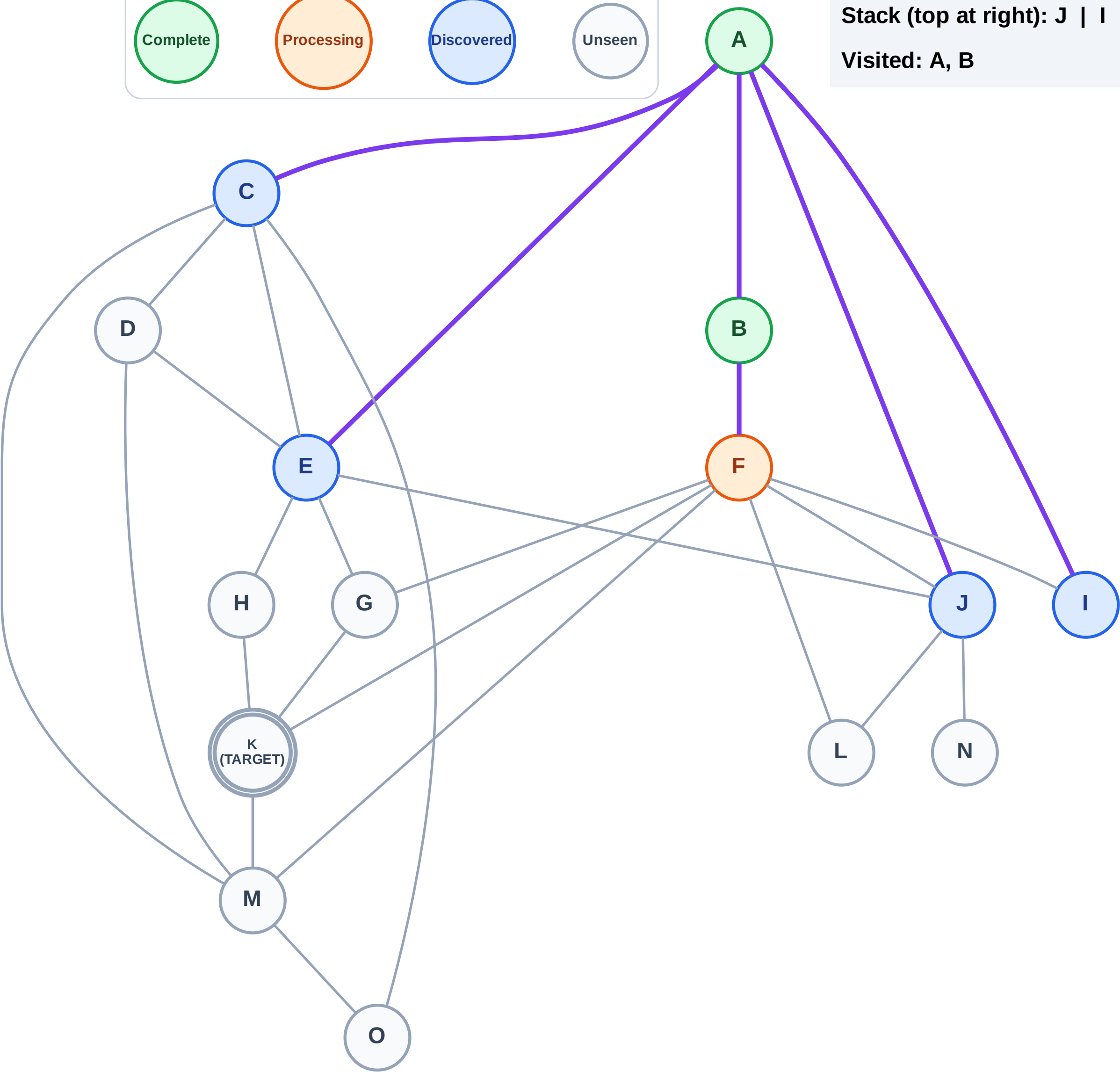
Processing

Discovered

Unseen

Stack (top at right): J | I | E | C

Visited: A, B



DFS Traversal

Step 7: Discover M, L, K, G from F

Legend



Complete



Processing



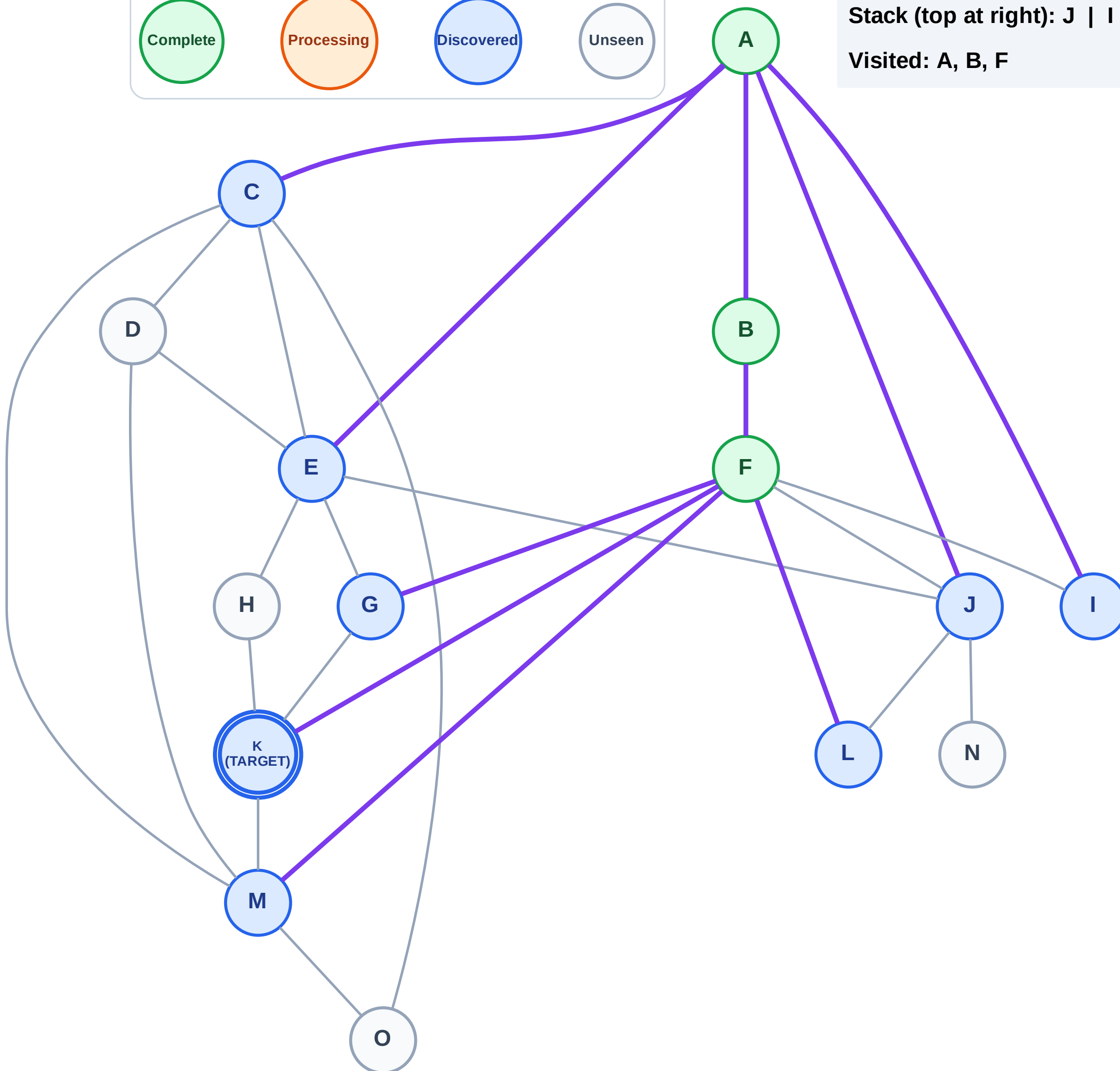
Discovered



Unseen

Stack (top at right): J | I | E | C | M | L | K | G

Visited: A, B, F



DFS Traversal

Step 8: Process node G

Legend

Complete

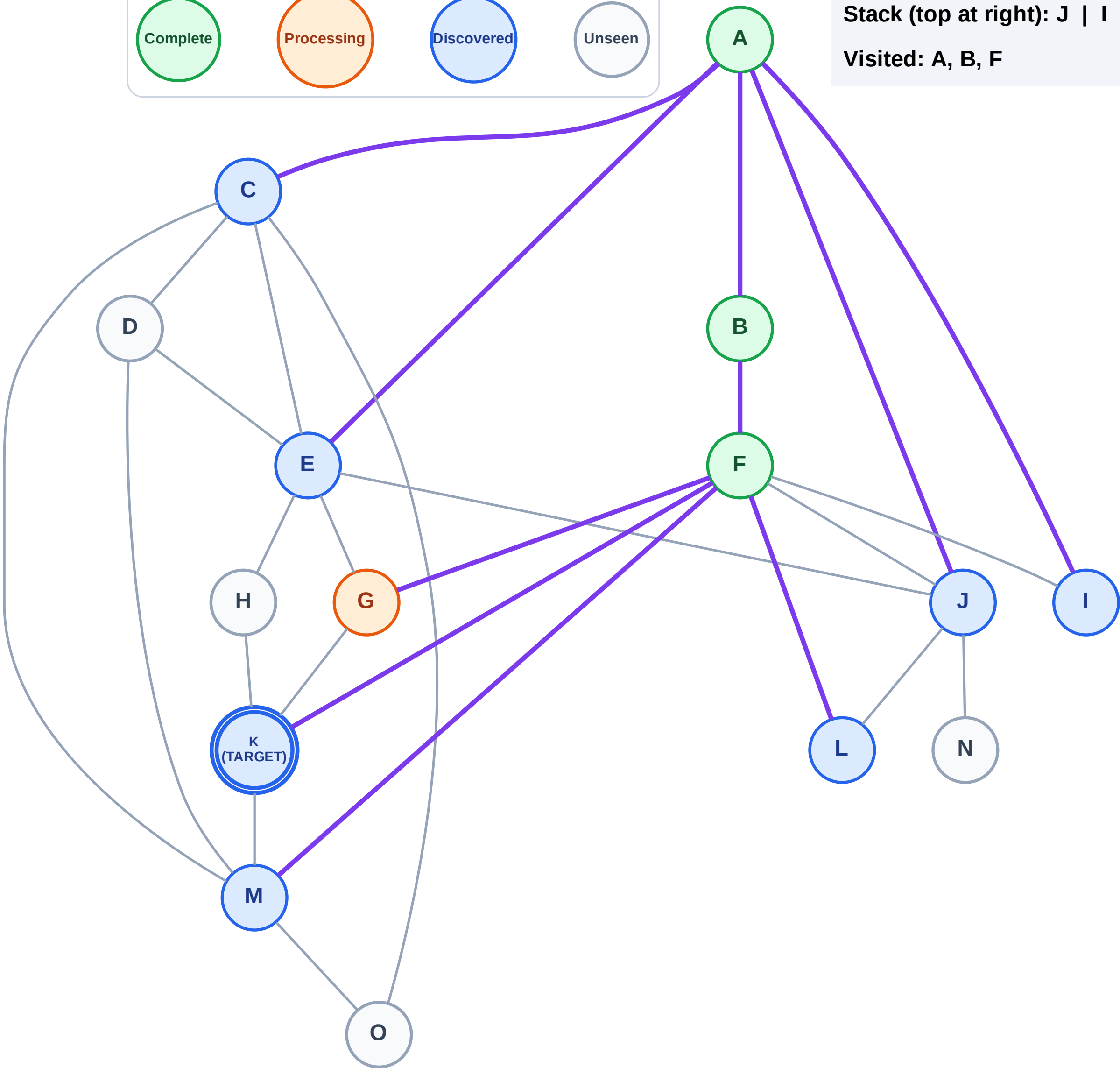
Processing

Discovered

Unseen

Stack (top at right): J | I | E | C | M | L | K

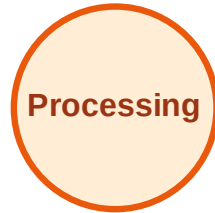
Visited: A, B, F



DFS Traversal

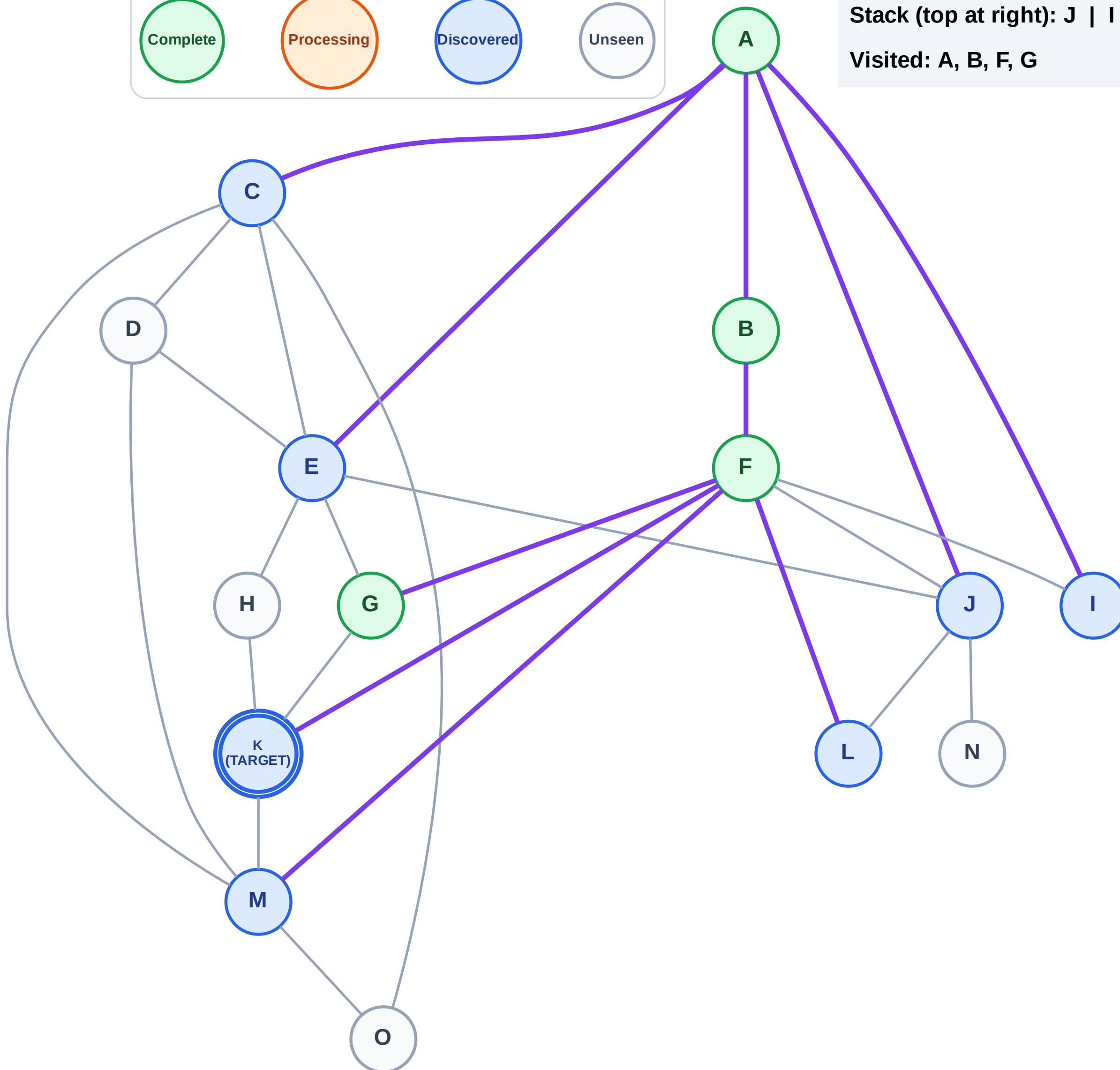
Step 9: No new neighbors from G

Legend



Stack (top at right): J | I | E | C | M | L | K

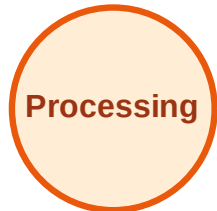
Visited: A, B, F, G



DFS Traversal

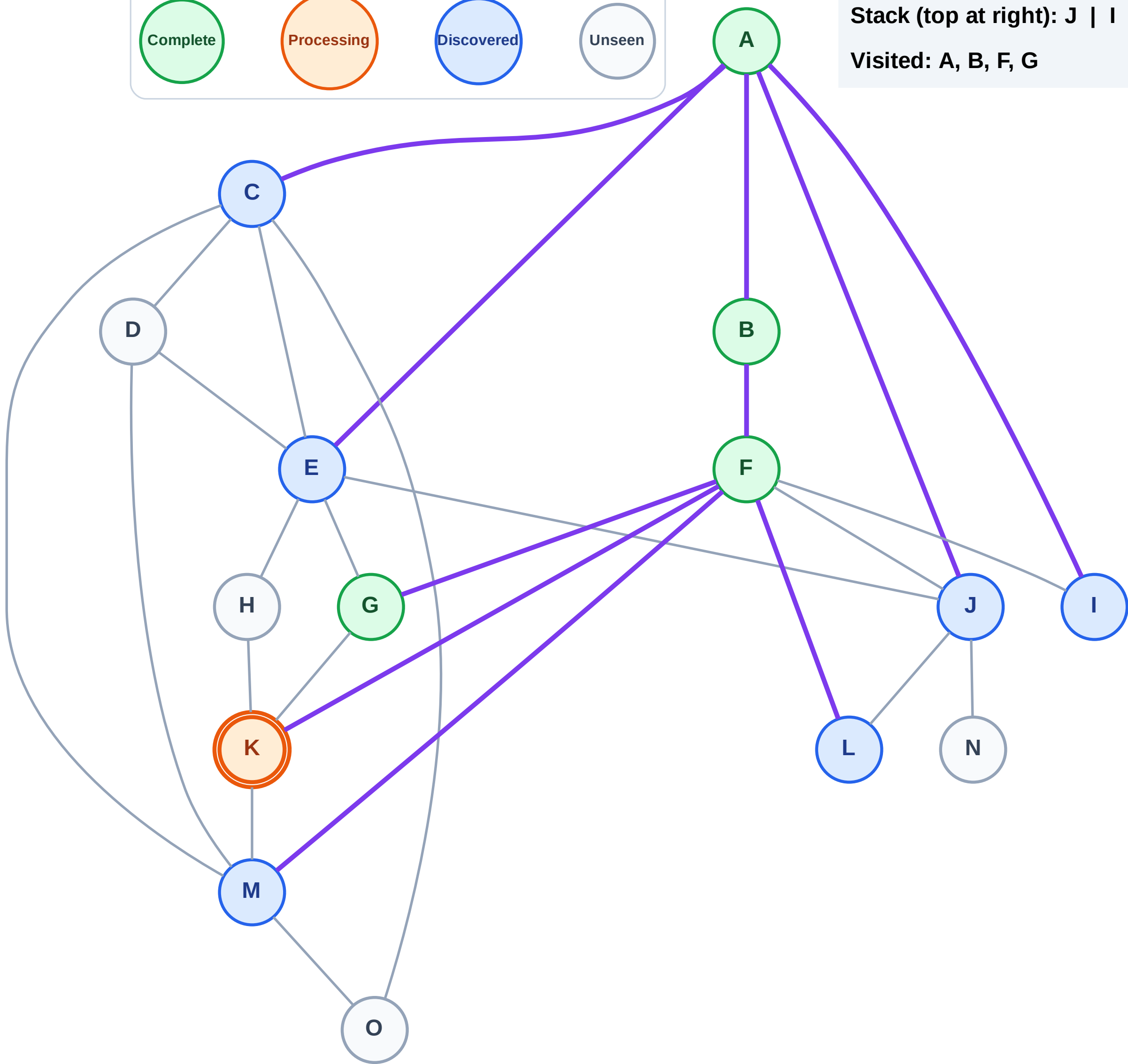
Step 10: Process node K

Legend



Stack (top at right): J | I | E | C | M | L

Visited: A, B, F, G



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): J | I | E | C | M | L

Visited: A, B, F, G, K

